

TMA Training Function

Mathematics for Machine Learning

<i>Course</i>	Mathematics for Machine Learning
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Objectives

- Introduce trainees to fundamental mathematical concepts essential for comprehending algorithms in machine learning and deep learning

Contents

1. Linear Algebra
2. Multivariable Calculus
3. Descriptive Statistics
4. Correlation and Regression
5. Probability and Probability Distribution

1. 1. Vector

1. A vector is a tuple of one or more values called scalars.
2. A vector as a list of numbers, and vector algebra as operations performed on the numbers in the list.
3. Vectors are used to
 1. Represent numbers for machines to process and transform inputs
 2. Represent the output of our ML model

1. 2. Matrices

- **Matrix** is a rectangular array of real-valued scalars arranged in m horizontal rows and n vertical columns
 - Each element a_{ij} belongs to the i^{th} row and j^{th} column
 - The elements are denoted a_{ij} or $\mathbf{A}_{ij}$ or $[\mathbf{A}]_{ij}$ or $\mathbf{A}(i, j)$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- For the matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the size (dimension) is $m \times n$ or (m, n)
 - Matrices are denoted by bold-font upper-case letters

1.3. Transpose of Matrices

- **Transpose** of the matrix: $\mathbf{A}^T$ has the rows and columns exchanged

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \Rightarrow \mathbf{x}^T = [x_1 \ x_2 \ \dots \ x_m]; \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \ddots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \Rightarrow \mathbf{A}^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \dots & \dots & \ddots & \dots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}$$

- Some properties

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$$

$$(\mathbf{A}^T)^T = \mathbf{A}$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- **Square matrix**: has the same number of rows and columns
- **Identity matrix** ($\mathbf{I}_n$): has ones on the main diagonal, and zeros elsewhere

- E.g.: identity matrix of size 3×3 :

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.4. Matrix Operators – Addition/subtraction

- Addition or subtraction

$$\begin{bmatrix} 1 & 3 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 5 \\ 7 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 1+0 & 3+0 & 1+5 \\ 1+7 & 0+5 & 0+0 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 6 \\ 8 & 5 & 0 \end{bmatrix}$$

- Scalar multiplication

$$(c\mathbf{A})_{i,j} = c \cdot \mathbf{A}_{i,j}$$

$$2 \cdot \begin{bmatrix} 1 & 8 & -3 \\ 4 & -2 & 5 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 8 & 2 \cdot -3 \\ 2 \cdot 4 & 2 \cdot -2 & 2 \cdot 5 \end{bmatrix} = \begin{bmatrix} 2 & 16 & -6 \\ 8 & -4 & 10 \end{bmatrix}$$

1.4. Matrix – Hadamard product

- Elementwise multiplication of two matrices **A** and **B** is called the *Hadamard product* or *elementwise product*
 - The math notation is $\odot$

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} & \dots & a_{1n}b_{1n} \\ a_{21}b_{21} & a_{22}b_{22} & \dots & a_{2n}b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}b_{m1} & a_{m2}b_{m2} & \dots & a_{mn}b_{mn} \end{bmatrix}$$

$$\begin{aligned} A \odot B &= B \odot A, \\ A \odot (B \odot C) &= (A \odot B) \odot C, \\ A \odot (B + C) &= A \odot B + A \odot C, \\ (kA) \odot B &= A \odot (kB) = k(A \odot B), \\ A \odot 0 &= 0 \odot A = 0. \end{aligned}$$

1.4. Matrix - Multiplication

- To multiply two matrices $\mathbf{A} \in \mathbb{R}^{n \times k}$ and $\mathbf{B} \in \mathbb{R}^{k \times m}$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix}$$

$\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}$

$$\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times p}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad \forall 1 \leq i \leq m, 1 \leq j \leq p$$

- We can consider the **matrix-matrix product** as dot-products of rows in $\mathbf{A}$ and columns in $\mathbf{B}$

$$\mathbf{AB} = \begin{pmatrix} a_{11}b_{11} + \cdots + a_{1n}b_{n1} & a_{11}b_{12} + \cdots + a_{1n}b_{n2} & \cdots & a_{11}b_{1p} + \cdots + a_{1n}b_{np} \\ a_{21}b_{11} + \cdots + a_{2n}b_{n1} & a_{21}b_{12} + \cdots + a_{2n}b_{n2} & \cdots & a_{21}b_{1p} + \cdots + a_{2n}b_{np} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}b_{11} + \cdots + a_{mn}b_{n1} & a_{m1}b_{12} + \cdots + a_{mn}b_{n2} & \cdots & a_{m1}b_{1p} + \cdots + a_{mn}b_{np} \end{pmatrix}$$

- Size: $\mathbf{A}(m \times n) \cdot \mathbf{B}(n \times p) = \mathbf{C}(m \times p)$

1.4. Matrix - Multiplication

■ Properties

- $AB \neq BA$
- $A(B+C) = AB + AC$
- $(A+B)C = AC + BC$
- $(AB)C = A(BC)$
- $cA = Ac$
- $c(AB) = (cA)B = A(cB)$
- $(AB)c = A(Bc)$
- $(AB)^T = B^T A^T$

1.5. Matrix – Linear Dependence

- For the following matrix $\mathbf{B} = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$
- Notice that for the two columns $\mathbf{b}_1 = [2, 4]^T$ and $\mathbf{b}_2 = [-1, -2]^T$, we can write $\mathbf{b}_1 = -2 \cdot \mathbf{b}_2$
 - This means that the two columns are linearly dependent
- The weighted sum $a_1 \mathbf{b}_1 + a_2 \mathbf{b}_2$ is referred to as a **linear combination** of the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$
 - In this case, a linear combination of the two vectors exist for which $\mathbf{b}_1 + 2 \cdot \mathbf{b}_2 = \mathbf{0}$
- A collection of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are **linearly dependent** if there exist coefficients $a_1, a_2, \dots, a_k$ not all equal to zero, so that

$$\sum_{i=1}^k a_i \mathbf{v}_i = \mathbf{0}$$

1.6. Matrix Rank

- For an $n \times m$ matrix, the *rank* of the matrix is the largest number of linearly independent columns
- The matrix **B** from the previous example has $rank(\mathbf{B}) = 1$, since the two columns are linearly dependent

$$\mathbf{B} = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$$

- The matrix **C** below has $rank(\mathbf{C}) = 2$, since it has two linearly independent columns
 - i.e., $\mathbf{c}_4 = -1 \cdot \mathbf{c}_1$, $\mathbf{c}_5 = -1 \cdot \mathbf{c}_3$, $\mathbf{c}_2 = 3 \cdot \mathbf{c}_1 + 3 \cdot \mathbf{c}_3$

$$\mathbf{C} = \begin{bmatrix} 1 & 3 & 0 & -1 & 0 \\ -1 & 0 & 1 & 1 & -1 \\ 0 & 3 & 1 & 0 & -1 \\ 2 & 3 & -1 & -2 & 1 \end{bmatrix}$$

1.6. Matrix Rank - properties

- $\text{rank}(A) = \text{rank}(A^T)$
- $\text{rank}(A) \leq \min(m, n), A \in R^{m \times n}$
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
- $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B).$
- $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB)$
 - $A \in R^{m \times n}, B \in R^{n \times p}$

1.6. Matrix - Determinant

- **Determinant** of a matrix, denoted by $\det(\mathbf{A})$ or $|\mathbf{A}|$, is a real-valued scalar encoding certain properties of the matrix

- E.g., for a matrix of size 2×2 : $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$

- For larger-size matrices the determinant of a matrix is calculated as

$$\det(\mathbf{A}) = \sum_j a_{ij}(-1)^{i+j} \det(\mathbf{A}_{(i,j)})$$

- In the above, $\mathbf{A}_{(i,j)}$ is a **minor** of the matrix obtained by removing the row and column associated with the indices i and j

$$\begin{aligned} \det A &= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \cdot (-1)^{1+1} \begin{vmatrix} e & f \\ h & i \end{vmatrix} + b \cdot (-1)^{1+2} \begin{vmatrix} d & f \\ g & i \end{vmatrix} \\ &+ c \cdot (-1)^{1+3} \begin{vmatrix} d & e \\ g & h \end{vmatrix} = a(ei - hf) - b(di - fg) + c(dh - eg) \\ &= (aei + bfg + cdh) - (ahf + bdi + ceg) \end{aligned}$$

1.6. Matrix - Determinant properties

- If we interchange any two rows (or columns), then sign of the determinant changes.
- If any two rows or any two columns in a determinant are identical (or proportional), then the value of the determinant is zero.
- $\det(A) = \det(A^T)$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = \frac{1}{\det(A)}$
 - $\det(A^{-1})\det(A) = \det(A^{-1}A) = \det(I) = 1$

1.7. Inverse of a Matrix

- For a square $n \times n$ matrix $\mathbf{A}$ with rank n , $\mathbf{A}^{-1}$ is its *inverse matrix* if their product is an identity matrix $\mathbf{I}$

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

- Properties of inverse matrices

$$(\mathbf{A}^{-1})^{-1} = \mathbf{A}$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

- If $\det(\mathbf{A}) = 0$ (i.e., $\text{rank}(\mathbf{A}) < n$), then the inverse does not exist
 - A matrix that is not invertible is called a *singular matrix*
- Note that finding an inverse of a large matrix is computationally expensive
 - In addition, it can lead to numerical instability
- If the inverse of a matrix is equal to its transpose, the matrix is said to be *orthogonal matrix*

$$\mathbf{A}^{-1} = \mathbf{A}^T$$

1.8. Inverse of a Matrix formula

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

- $A^{-1} = \frac{1}{\det(A)} [C_{ij}]^T$
 - $C_{ij} = (-1)^{i+j} \det(M_{ij})$
 - Where M_{ij} refers to the $(i, j)^{\text{th}}$ minor matrix when i^{th} row and j^{th} column is removed.

1.9. Inverse of a Matrix formula example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Substituting the values from matrix A , we get:

$$\det(A) = 1 \times (1 \times 0 - 4 \times 6) - 2 \times (0 \times 0 - 4 \times 5) + 3 \times (0 \times 6 - 1 \times 5)$$

$$\det(A) = 1 \times (0 - 24) - 2 \times (0 - 20) + 3 \times (0 - 5)$$

$$\det(A) = 1 \times (-24) - 2 \times (-20) + 3 \times (-5)$$

$$\det(A) = -24 + 40 - 15 = 1$$

1.9. Inverse of a Matrix formula example

$$\text{Cofactor}(A) = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix} \quad A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

where each C_{ij} will be calculated using the above formula.

$$C_{11} = (-1)^{1+1} \cdot \text{minor}_{11} = \det \begin{pmatrix} 1 & 4 \\ 6 & 0 \end{pmatrix} = 1 \times (-24 - 0 \times 6) = -24$$

$$C_{12} = (-1)^{1+2} \cdot \text{minor}_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 0 \end{pmatrix} = -1 \times (0 - 20) = 20$$

$$C_{13} = (-1)^{1+3} \cdot \text{minor}_{13} = \det \begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix} = 1 \times (0 - 5) = -5$$

$$C_{21} = (-1)^{2+1} \cdot \text{minor}_{21} = \det \begin{pmatrix} 2 & 3 \\ 6 & 0 \end{pmatrix} = -1 \times (0 - 18) = 18$$

1.9. Inverse of a Matrix formula example

$$C_{22} = (-1)^{2+2} \cdot \text{minor}_{22} = \det \begin{pmatrix} 1 & 3 \\ 5 & 0 \end{pmatrix} = 1 \times (0 - 15) = -15$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

$$C_{23} = (-1)^{2+3} \cdot \text{minor}_{23} = \det \begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix} = -1 \times (6 - 10) = 4$$

$$C_{31} = (-1)^{3+1} \cdot \text{minor}_{31} = \det \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} = 1 \times (8 - 3) = 5$$

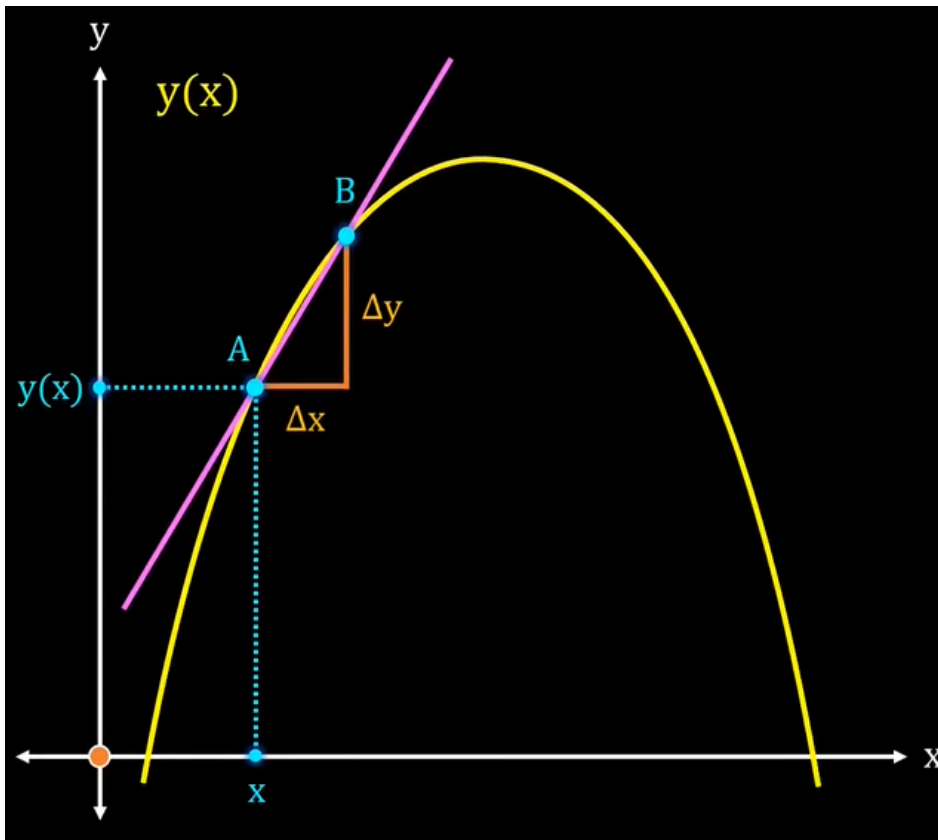
$$C_{32} = (-1)^{3+2} \cdot \text{minor}_{32} = \det \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix} = -1 \times (4 - 0) = -4$$

$$C_{33} = (-1)^{3+3} \cdot \text{minor}_{33} = \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1 \times (1 - 0) = 1$$

$$A^{-1} = \frac{1}{\det(A)} [C_{ij}]^T = \begin{pmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{pmatrix}^T = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$$

2.1. Derivative

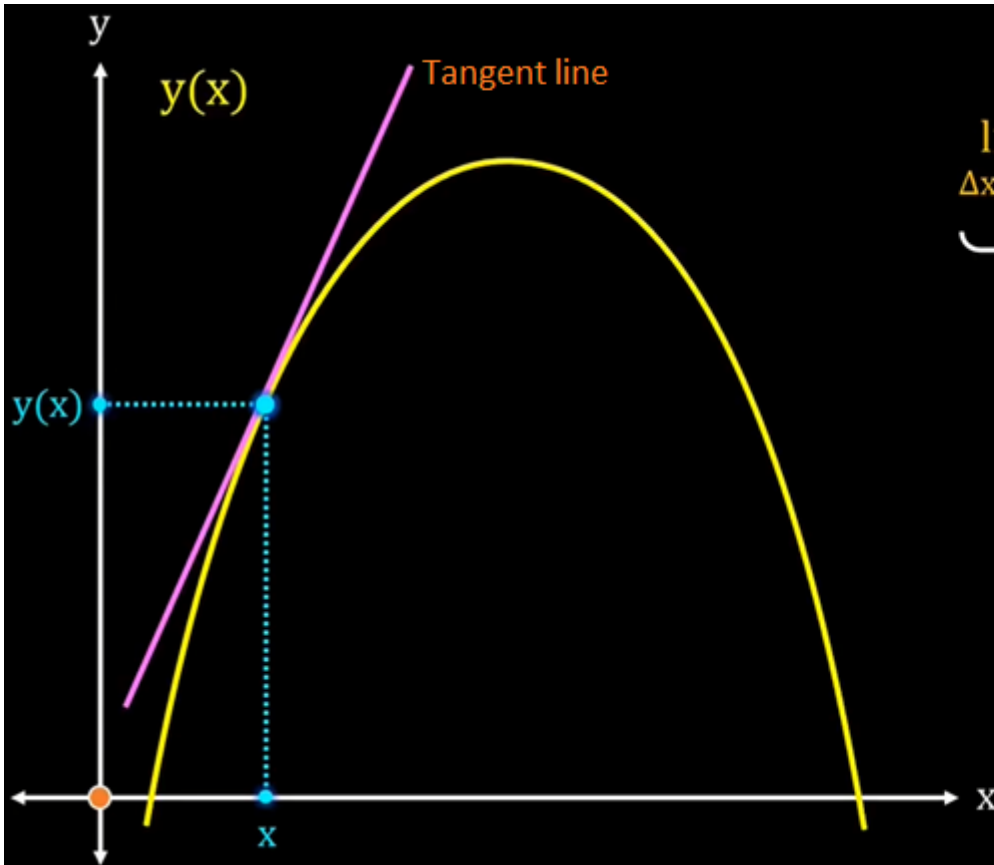
- Is the rate of change of a quantity y with respect to another quantity x . It is also termed the differential coefficient of y with respect to x



$$a = \frac{\Delta y}{\Delta x} = \frac{y(x + \Delta x) - y(x)}{\Delta x}$$

$\Delta x \rightarrow 0$

2.1. Derivative



$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x}$$

Slope of the tangent line at coordinates $(x, y(x))$

2.1. Derivative

$$y(x) = x^3$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{(x+\Delta x)^3 - x^3}{\Delta x}$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3 - x^3}{\Delta x}$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3}{\Delta x}$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{3x^2 + 3x\Delta x + (\Delta x)^2}{1}$$

$$y'(x) = 3x^2$$

2.1. Derivative

$$y(x) = \sin(x)$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{\sin(x+\Delta x) - \sin(x)}{\Delta x}$$

$$y'(x) = \lim_{\Delta x \rightarrow 0} \frac{\sin(x)\cos(\Delta x) + \sin(\Delta x)\cos(x) - \sin(x)}{\Delta x}$$

$$y'(x) = \sin(x) \lim_{\Delta x \rightarrow 0} \frac{\cos(\Delta x) - 1}{\Delta x} + \cos(x) \lim_{\Delta x \rightarrow 0} \frac{\sin(\Delta x)}{\Delta x}$$

$$y'(x) = \sin(x) * 0 + \cos(x) * 1$$

$$y'(x) = \cos(x)$$

2.1. Derivative - Properties

1. Linearity:

- The derivative of a sum of functions is the sum of the derivatives of the individual functions:

$$\frac{d}{dx}(f(x) + g(x)) = \frac{df}{dx} + \frac{dg}{dx}$$

- The derivative of a constant times a function is the constant times the derivative of the function:

$$\frac{d}{dx}(cf(x)) = c\frac{df}{dx}$$

2. Product Rule:

- The derivative of the product of two functions is the derivative of the first function times the second function plus the first function times the derivative of the second function:

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

3. Chain Rule:

- The chain rule allows us to differentiate composite functions. If $y = f(g(x))$, then:

$$\frac{dy}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$$

2.1. Derivative - Properties

4. Quotient Rule:

- The derivative of the quotient of two functions is the derivative of the numerator times the denominator minus the numerator times the derivative of the denominator, divided by the square of the denominator:

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

5. Power Rule:

- The derivative of a power of x is obtained by multiplying the exponent by the coefficient and decreasing the exponent by one:

$$\frac{d}{dx} (x^n) = nx^{n-1}$$

6. Derivative of a Constant:

- The derivative of a constant is zero:

$$\frac{d}{dx} (c) = 0$$

7. Derivative of the Identity Function:

- The derivative of the identity function $f(x) = x$ is one:

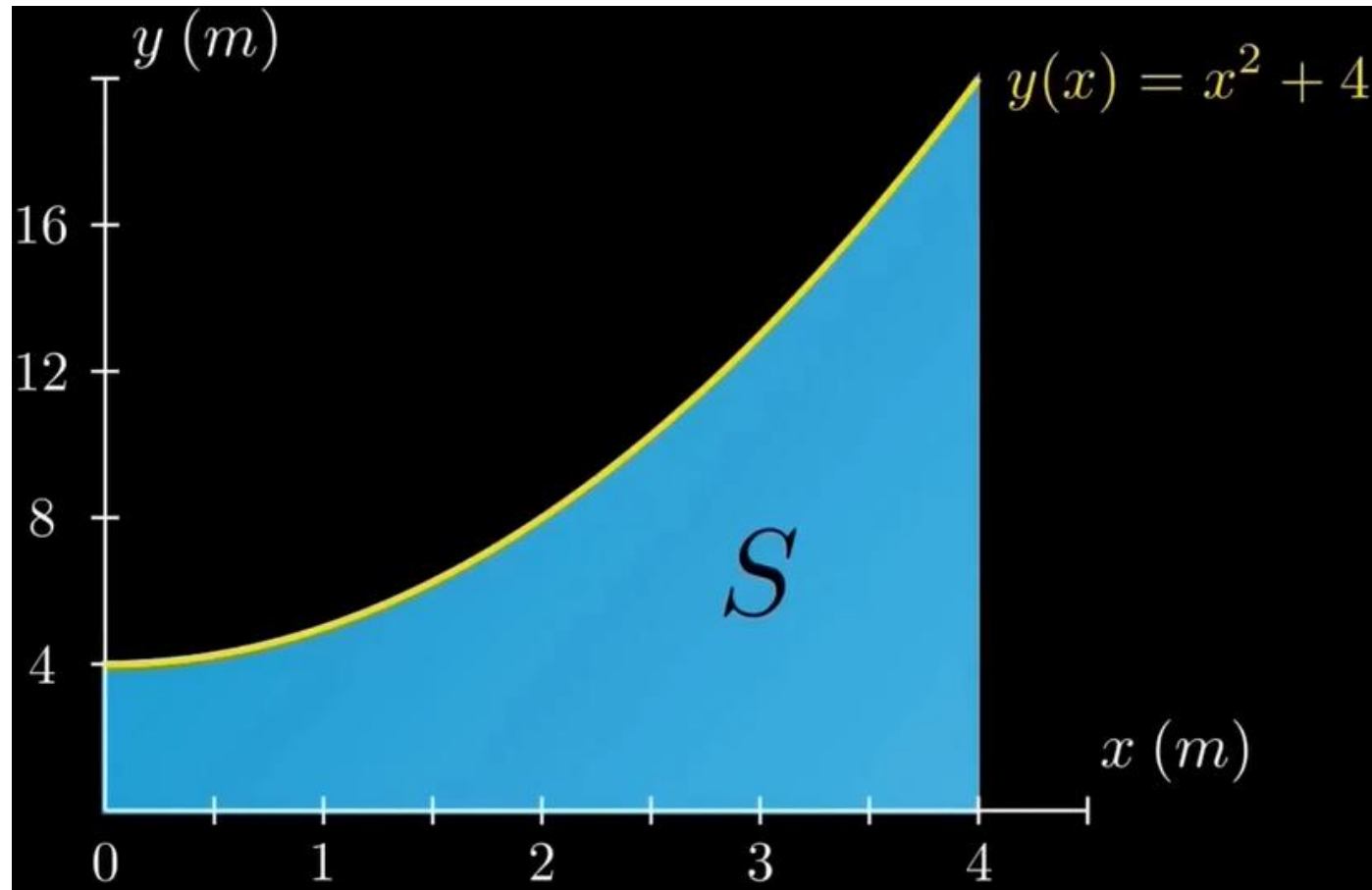
$$\frac{d}{dx} (x) = 1$$

2.2. Integral Calculus

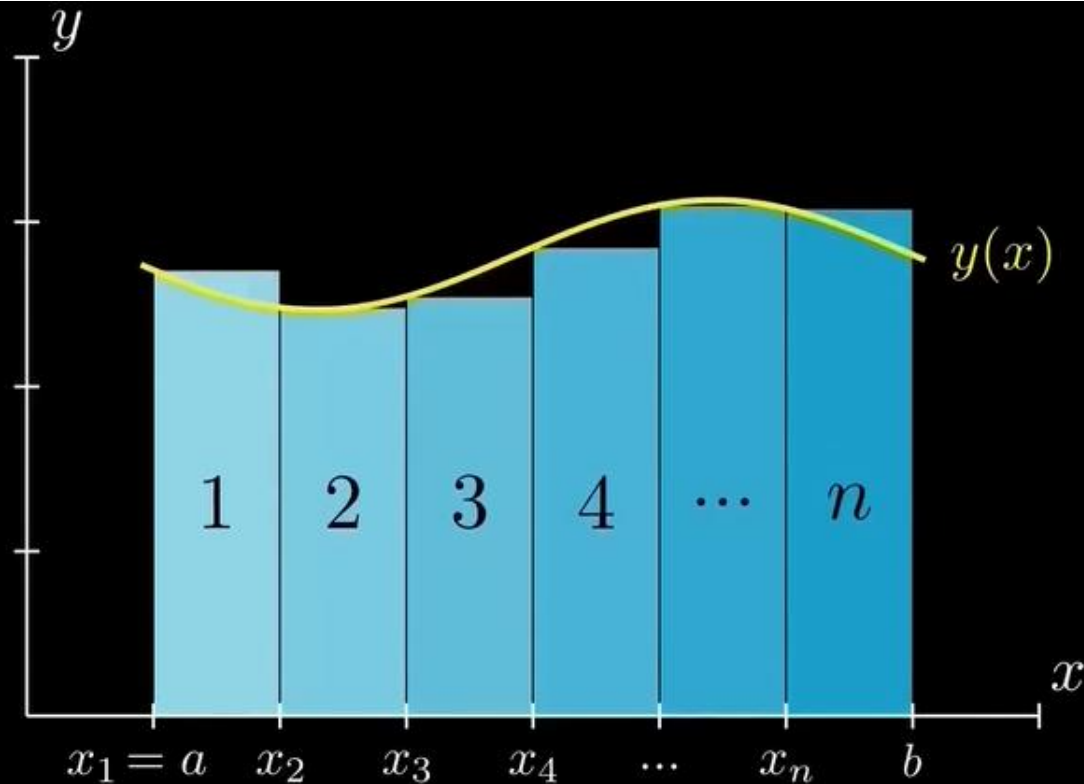
- Given a function $f(x)$ that is continuous on the interval $[a,b]$ we divide the interval into n subintervals of equal width, Δx , and from each interval choose a point, x_i . Then the **definite integral of $f(x)$ from a to b** is

- $$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i)\Delta x$$

2.2. Integral Calculus



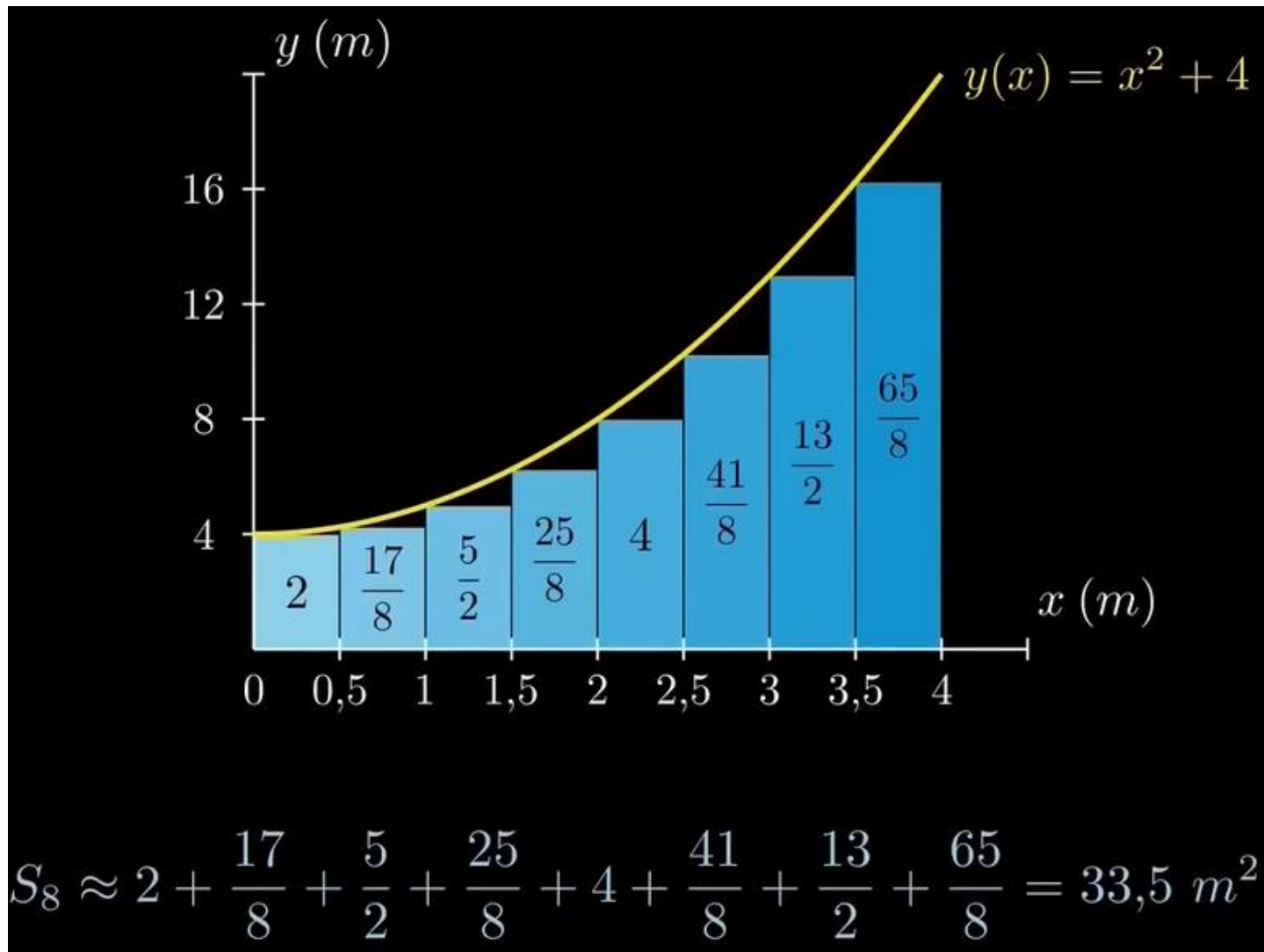
2.2. Integral Calculus



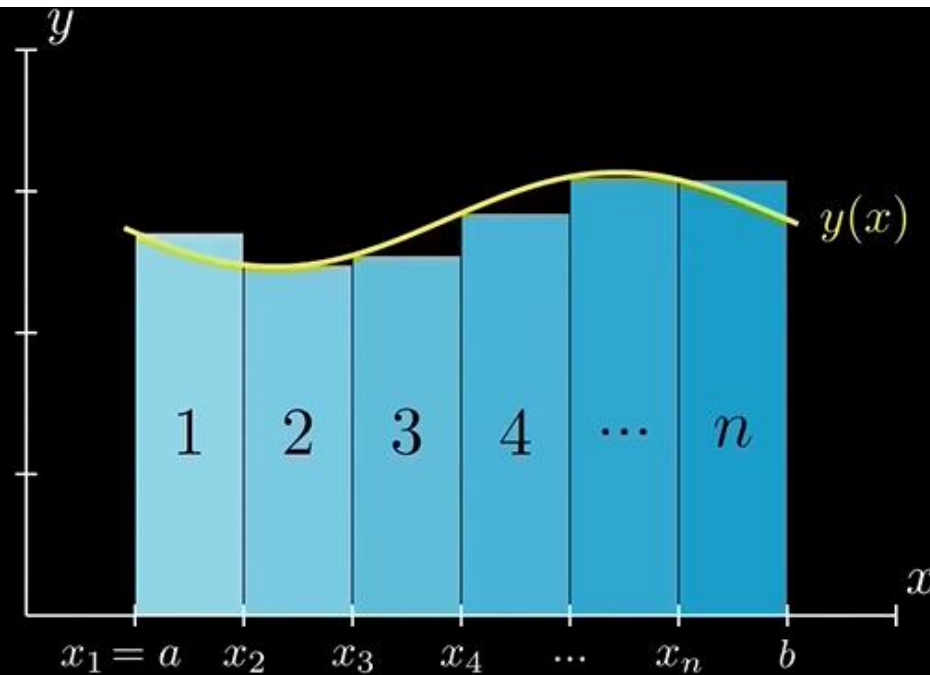
$$\Delta x = \frac{b - a}{n}$$

$$S \approx y(x_1)\Delta x + y(x_2)\Delta x + \dots + y(x_n)\Delta x \approx \sum_{i=1}^n y(x_i)\Delta x$$

2.2. Integral Calculus

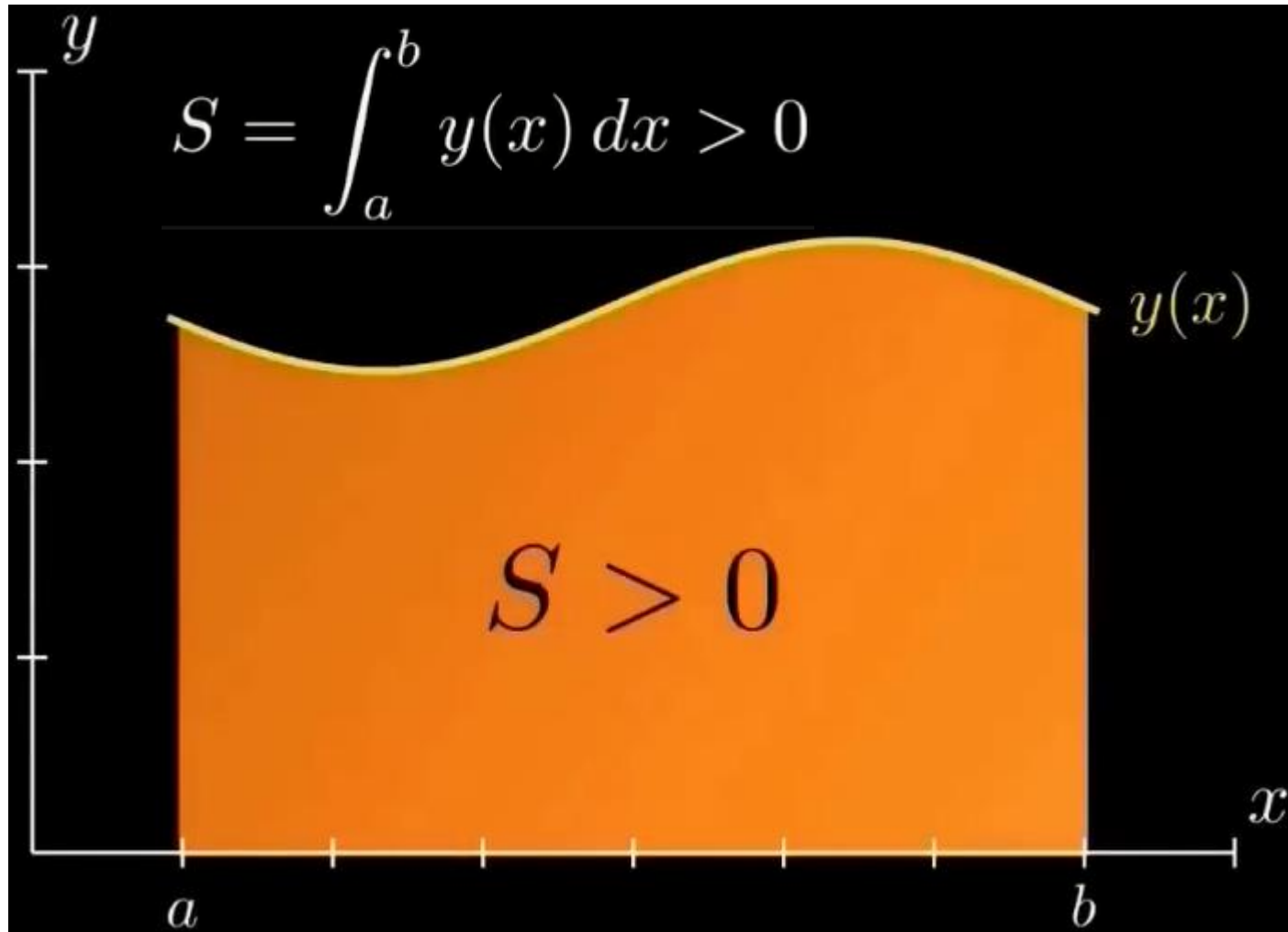


2.2. Integral Calculus

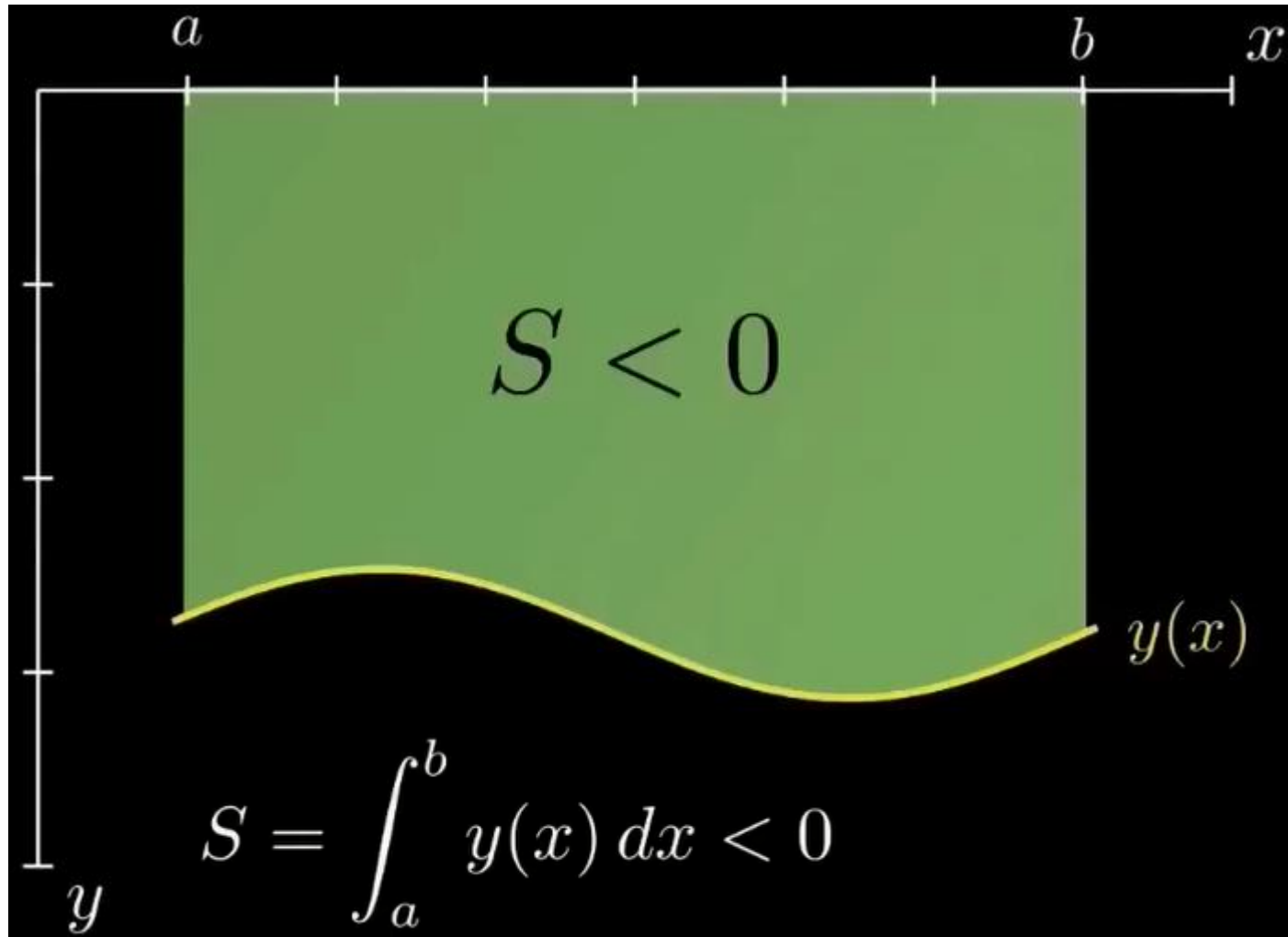


$$S \approx \sum_{i=1}^n y(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n y(x_i) \left(\frac{b-a}{n} \right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n y(x_i) \Delta x = \int_a^b y(x) dx$$

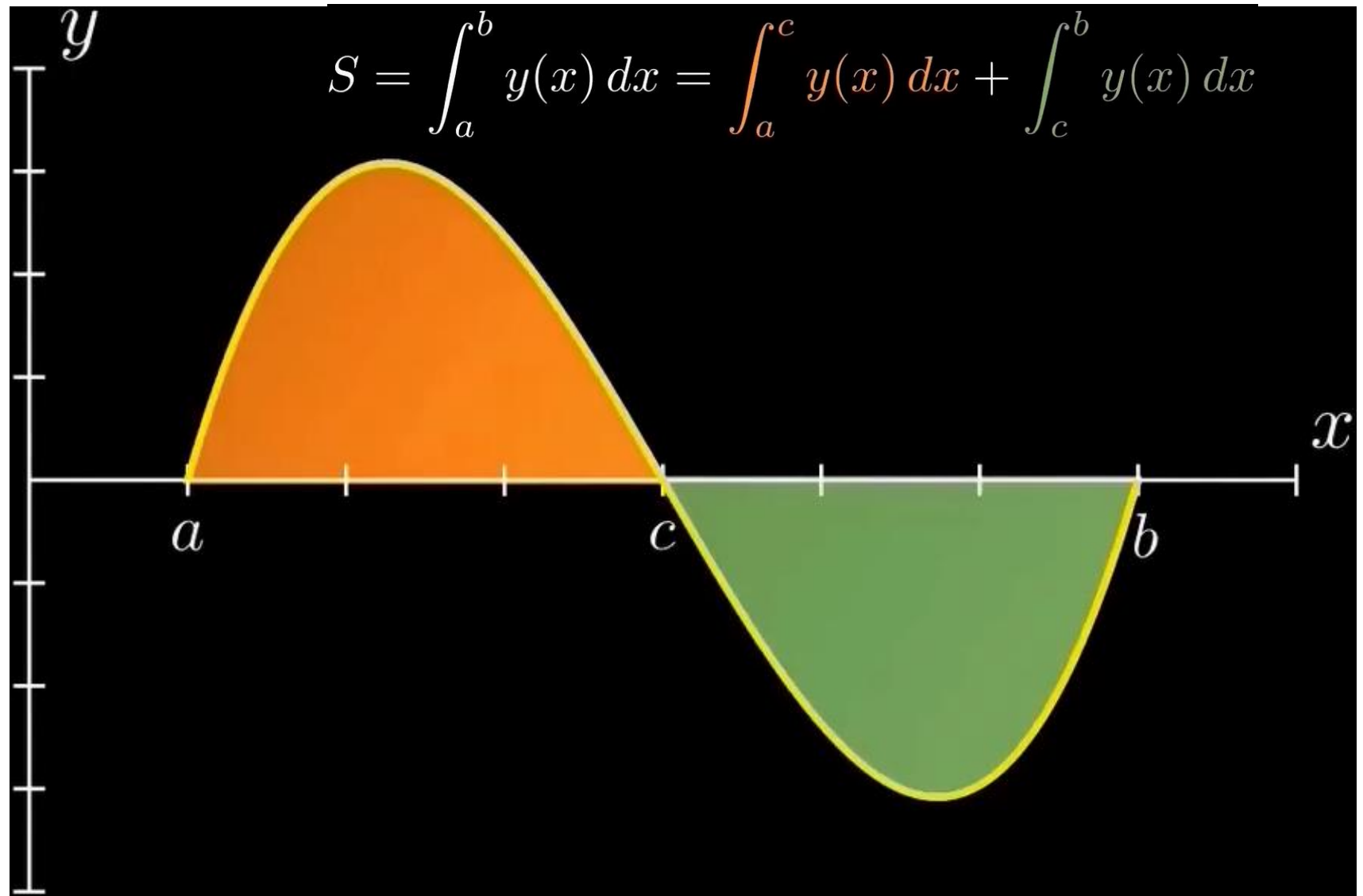
2.2. Integral Calculus



2.2. Integral Calculus



2.2. Integral Calculus



2.2. Integral Calculus - Properties

- $\int_a^b f(x)dx = -\int_b^a f(x)dx$
- $\int_a^a f(x)dx = 0$
- $\int_a^b cf(x)dx = c \int_a^b f(x)dx$
- $\int_a^b (f(x) \pm g(x))dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
- $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

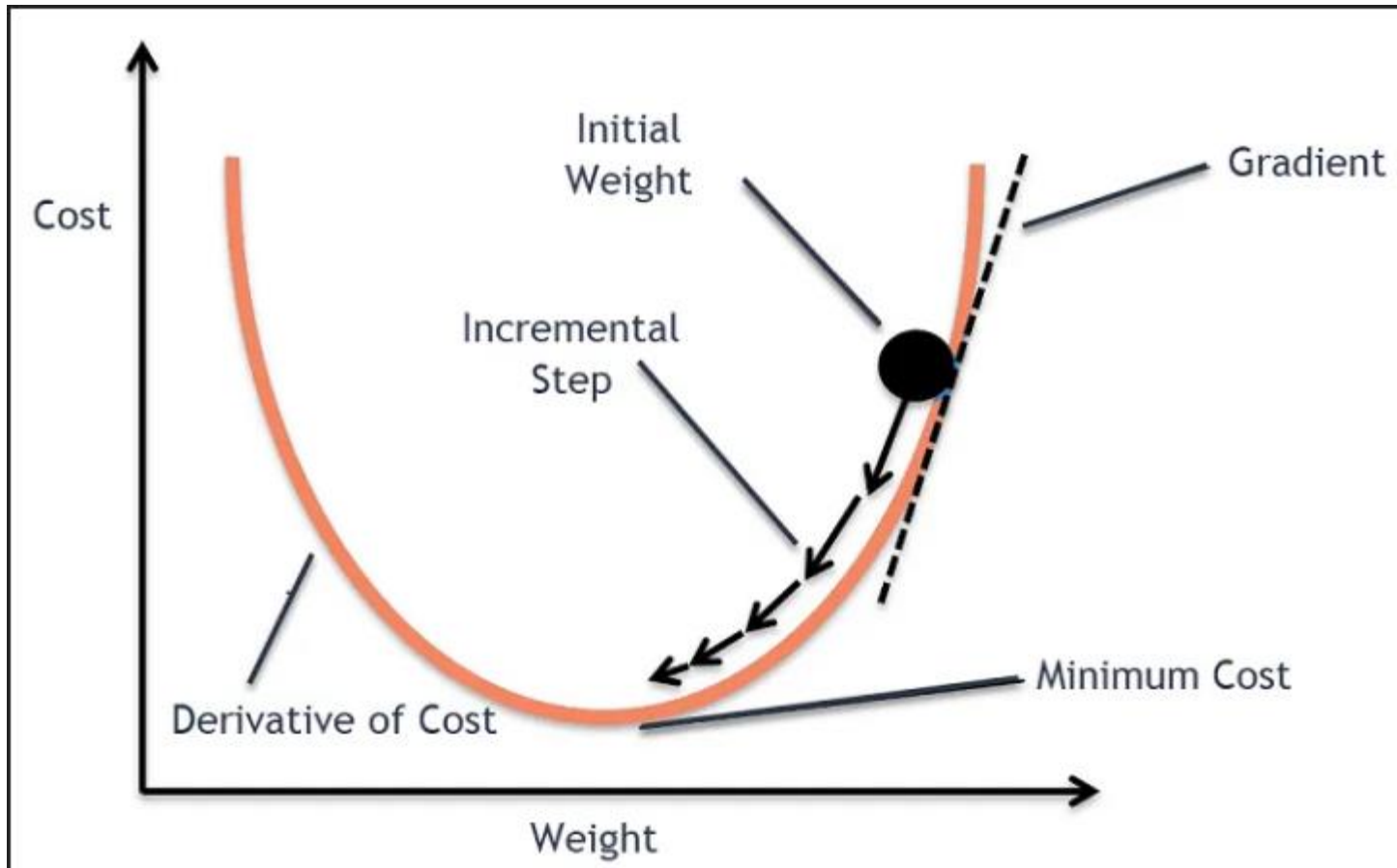
2.2. Integral Calculus - Properties

- $F(x)$ is the antiderivative of $f(x)$ if
 - $\frac{d}{dx}(F(x)) = f(x)$
- $\int f(x)dx = F(x) + c$
 - $F(x)$ is the antiderivative of $f(x)$.
 - dx is the differential element of the variable x .
 - c is the constant of integration
- $\int_a^b f(x)dx = F(b) - F(a)$
 - $f(x) = x^2, F(x) = \frac{x^3}{3}$
 - $\int_1^3 f(x)dx = F(3) - F(1) = \frac{3^3}{3} - \frac{1^3}{3} = \frac{26}{3}$

2.3.1 Gradient Descent

- To find out the values of a function's parameters (coefficients) that minimize a cost function as far as possible.
- It trains machine learning models by minimizing errors between predicted and actual results

2.3.2 Gradient Descent



2.3.3 Gradient Descent

- Function of one variable The Gradient Descent algorithm finds the minimum of the function by initializing the value of x at a random position, then moving x in the opposite direction of the derivative of $f(x)$. This operation is repeated until a certain threshold is reached.

- $x_{t+1} = x_t - \alpha * f'(x_t)$

2.3.4 Gradient Descent

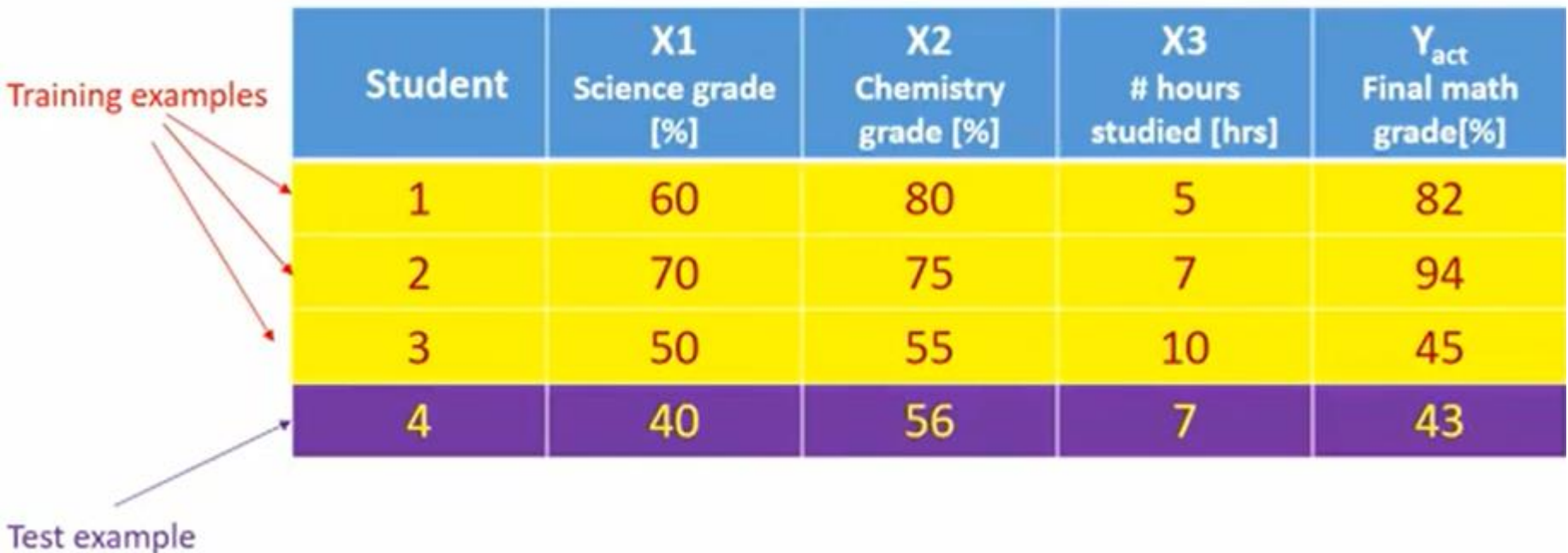
- Function of multiple variables: This algorithm calculates the gradient (derivative vector) of the function f at a random point x , then moves $\mathbf{x}$ in the opposite direction of this gradient.
 - $x_{t+1} = x_t - \alpha * \nabla f(x_t)$

2.4.1 Forward Propagation

- To find the actual output of neural networks. The input is fed to the network in a forward direction. It helps us find the actual output of each neuron.

2.4.2 Forward Propagation

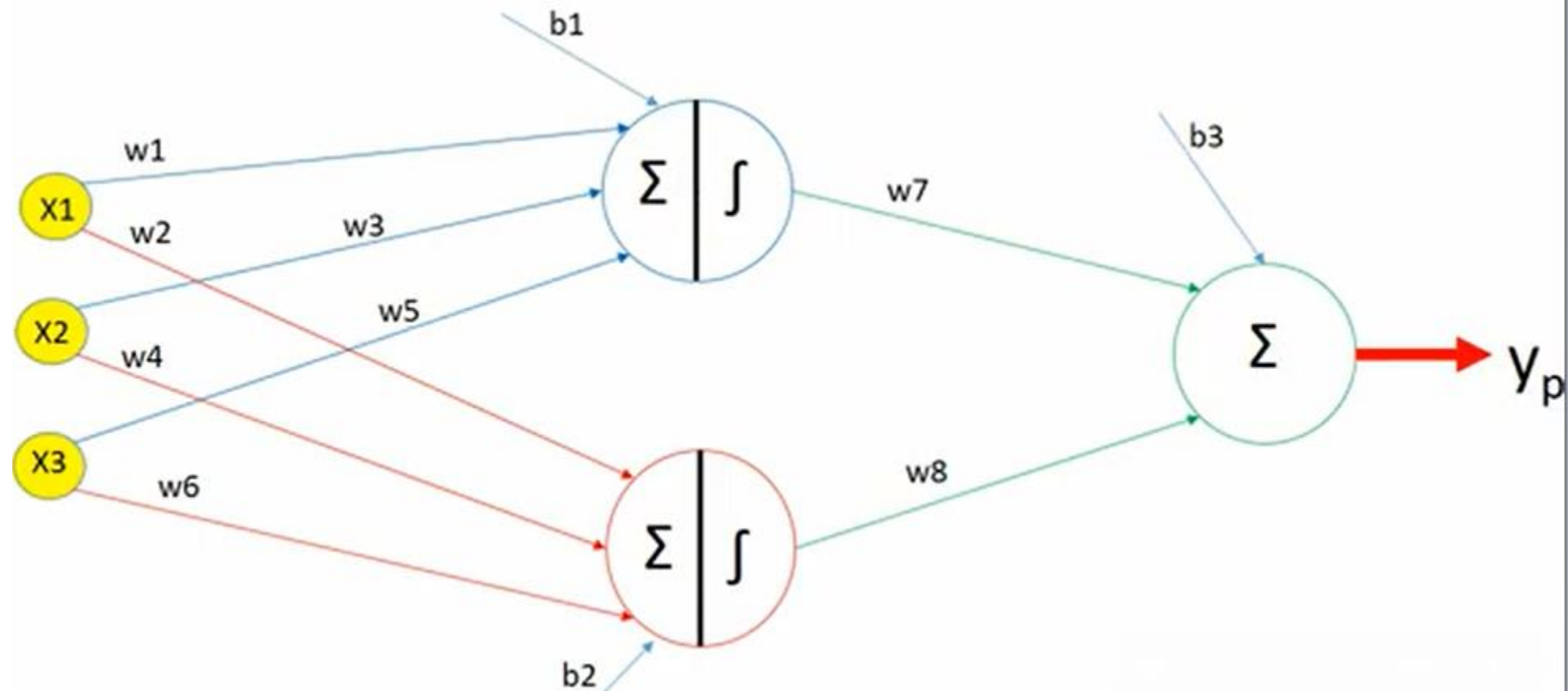
The Silly Training Dataset



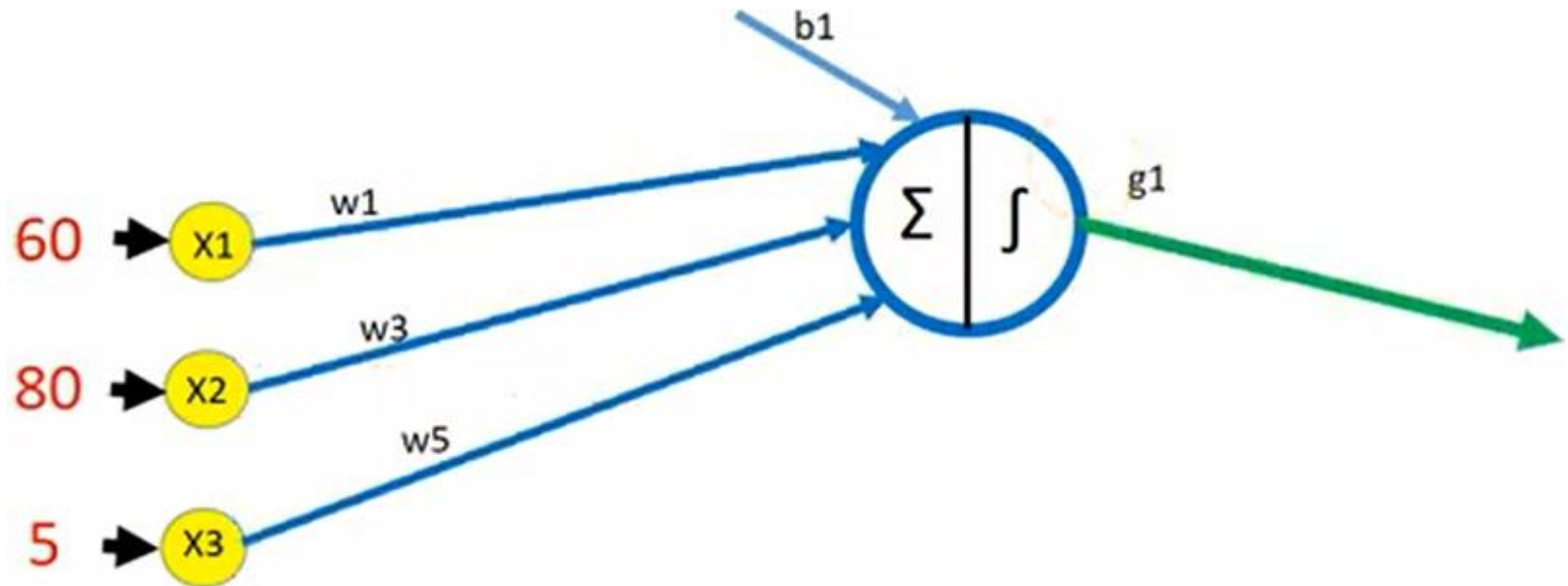
The diagram shows a table with five columns: Student, X1 Science grade [%], X2 Chemistry grade [%], X3 # hours studied [hrs], and Y_{act} Final math grade[%]. The first three rows are highlighted in yellow and labeled 'Training examples' with red arrows. The fourth row is highlighted in purple and labeled 'Test example' with a grey arrow.

Student	X1 Science grade [%]	X2 Chemistry grade [%]	X3 # hours studied [hrs]	Y _{act} Final math grade[%]
1	60	80	5	82
2	70	75	7	94
3	50	55	10	45
4	40	56	7	43

2.4.3 Forward Propagation



2.4.4 Forward Propagation



2.4.5 Forward Propagation

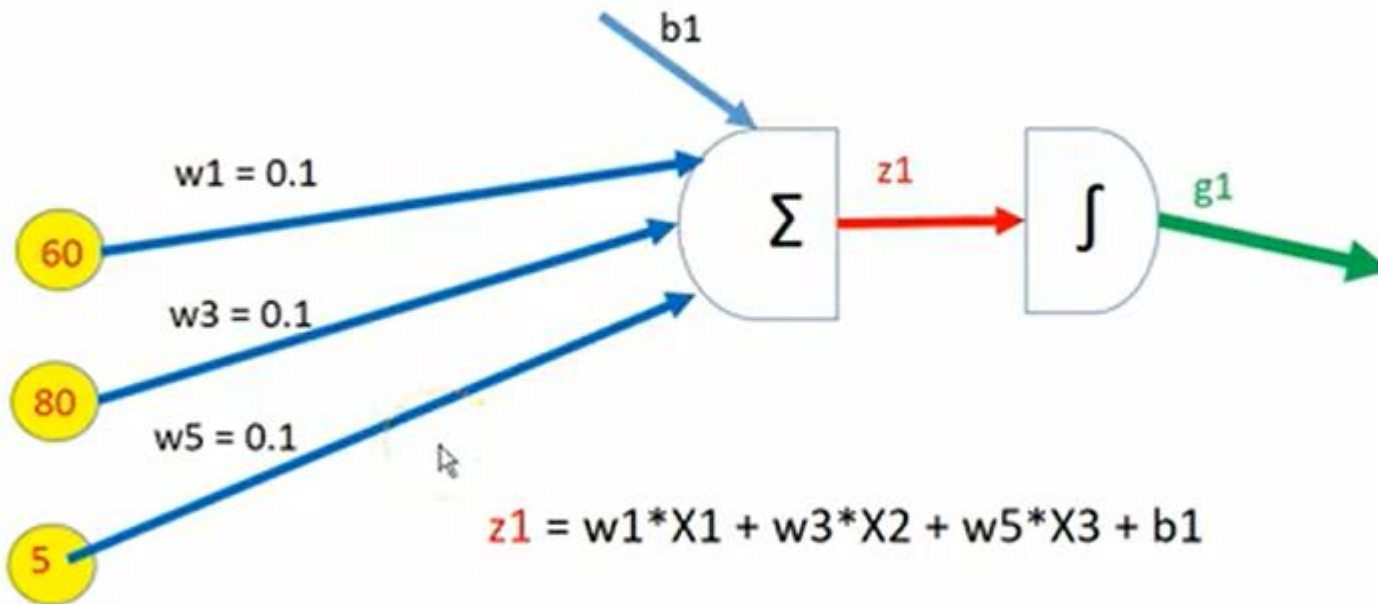
Inside the Neuron/Node

Two operations:

1. **Linear** summation of the weighted inputs, plus a bias
2. **Non-linear** operation

2.4.6 Forward Propagation

Linear Operation



$$z1 = w1 * X1 + w3 * X2 + w5 * X3 + b1$$

$$= 0.1 * 60 + 0.1 * 80 + 0.1 * 5 + (-15)$$

$$= -0.5$$

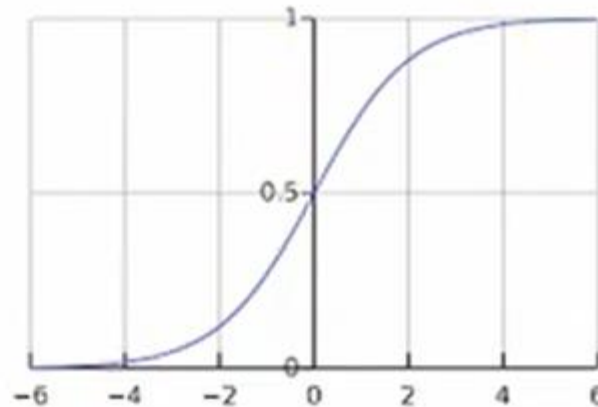
2.4.7 Forward Propagation

Non-linear Activation Function

For this example: **Sigmoid** function

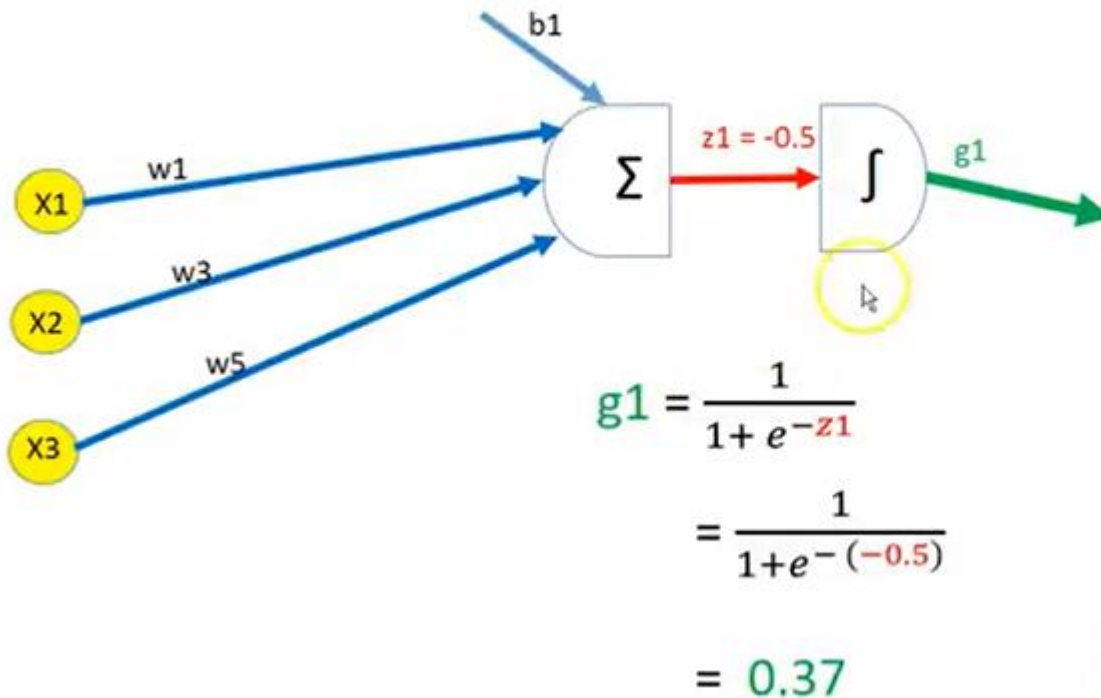
$$g1 = \frac{1}{1 + e^{-z1}}$$

14

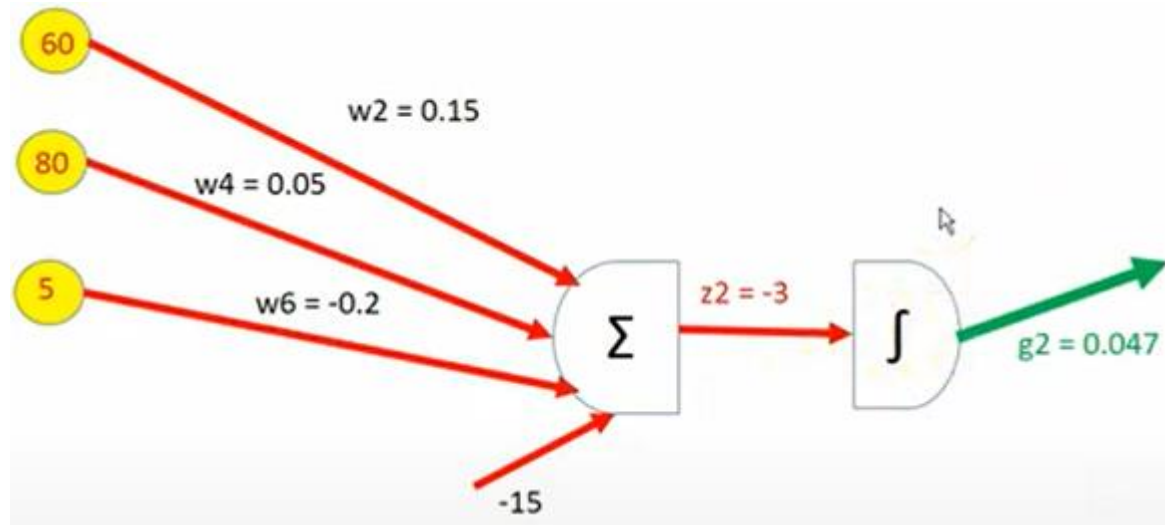


2.4.8 Forward Propagation

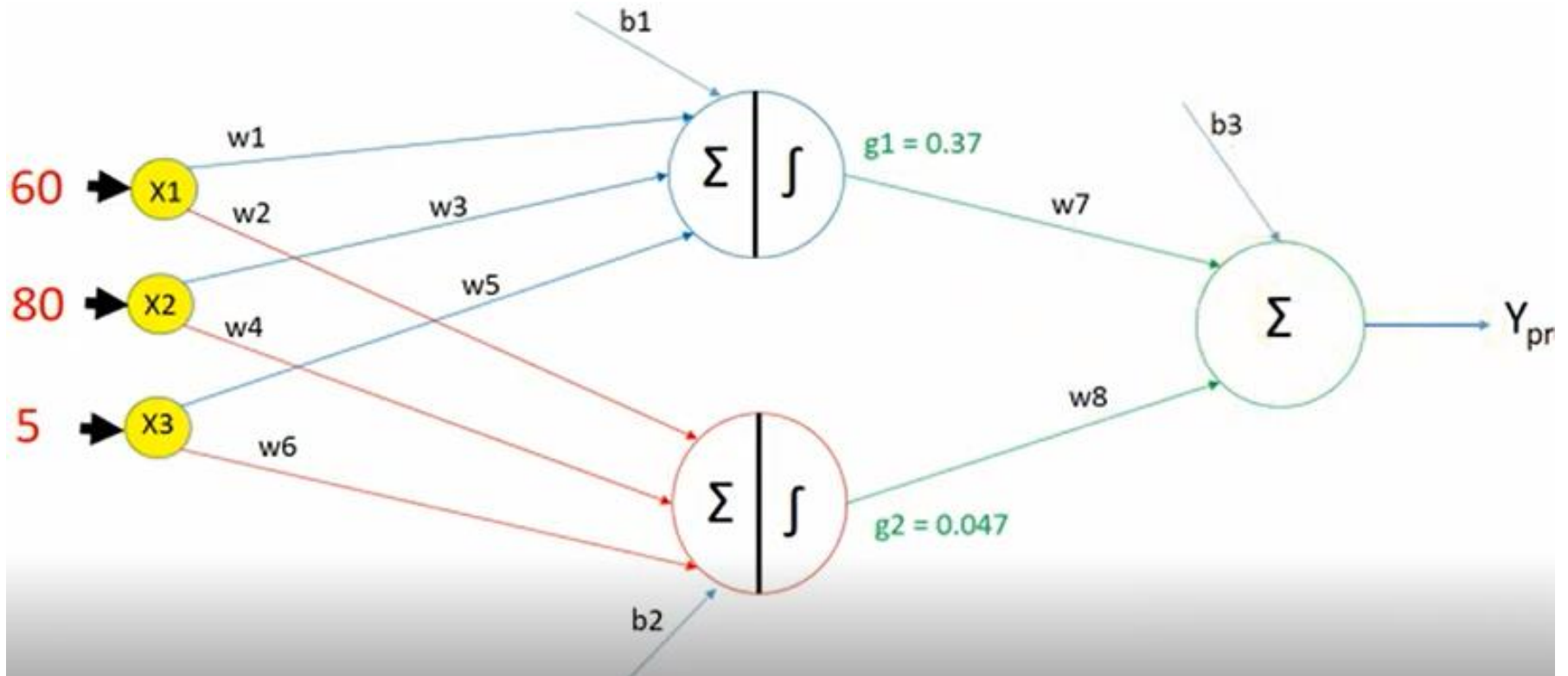
Non-Linear Operation



2.4.9 Forward Propagation

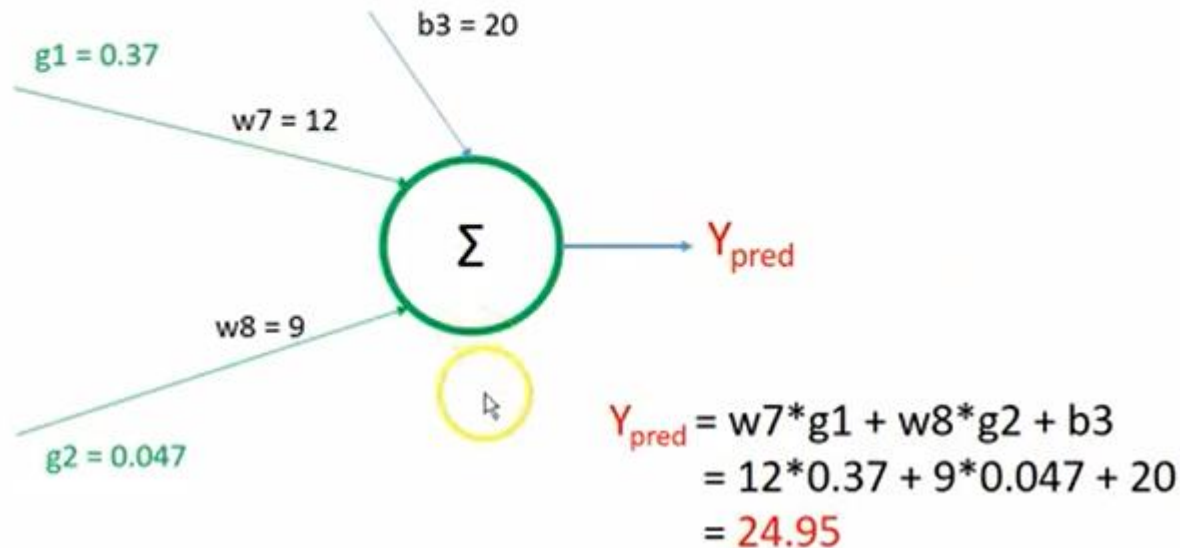


2.4.10 Forward Propagation

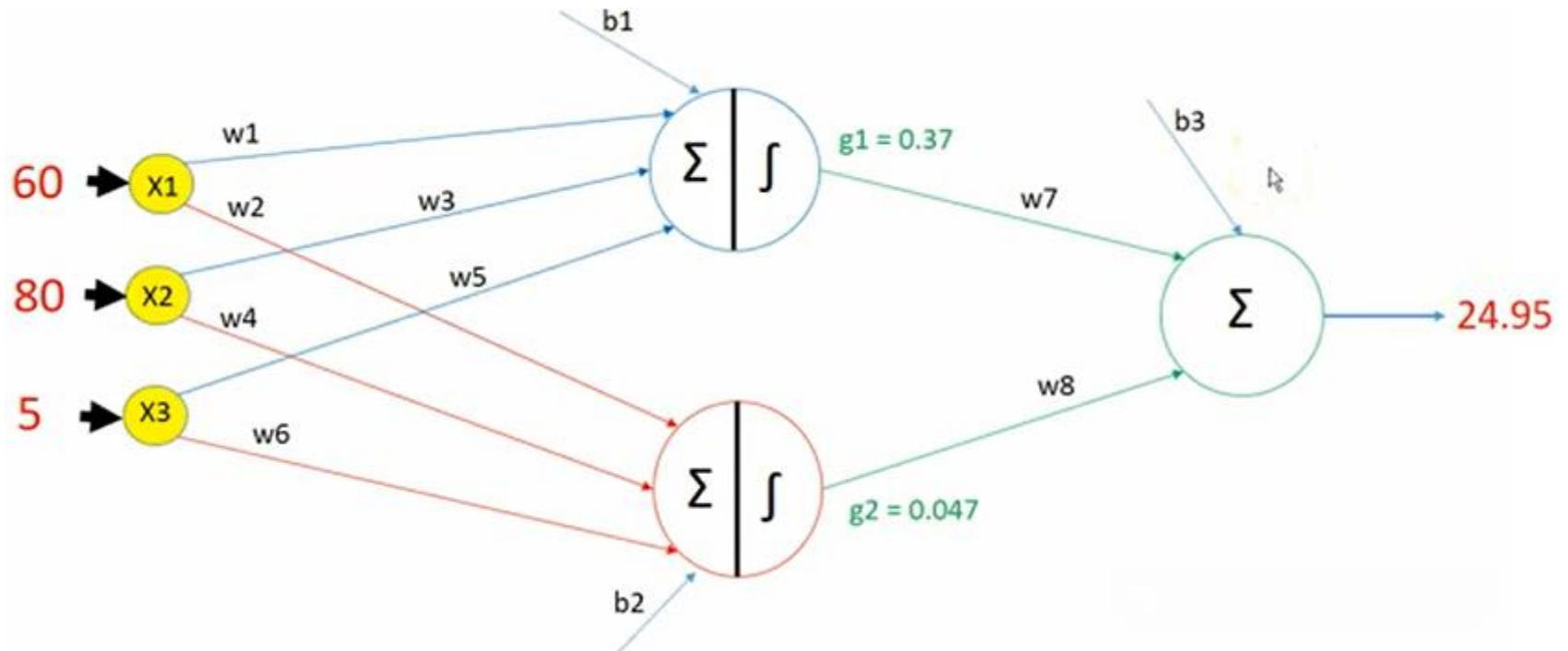


2.4.11 Forward Propagation

Output Layer: Linear Summation



2.4.12 Forward Propagation



2.5.1 Backpropagation

- We update the weights and biases until the error between the predicted output and the actual output is a minimum
- The best way is to perform Gradient Descent
 - Move along the negative direction of the slope of an error (cost) function until we find a minimum value
 - Calculate the partial derivatives of a cost function with respect to each parameter (weight or bias) then move in the negative direction of these derivatives (slopes)

2.5.2 Backpropagation

- We update the weights and biases until the error between the predicted output and the actual output is a minimum
- The best way is to perform Gradient Descent
 - Move along the negative direction of the slope of an error (cost) function until we find a minimum value
 - Calculate the partial derivatives of a cost function with respect to each parameter (weight or bias) then move in the negative direction of these derivatives (slopes)

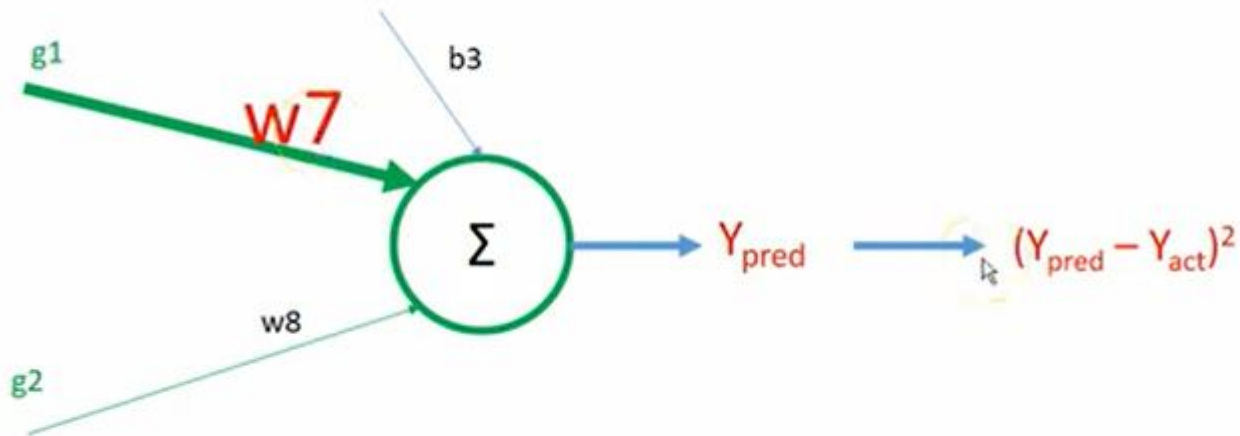
2.5.3 Backpropagation

- Select a cost function that calculates the error between predicted and actual

$$\text{Cost} = (Y_{\text{pred}} - Y_{\text{act}})^2$$

2.5.4 Backpropagation

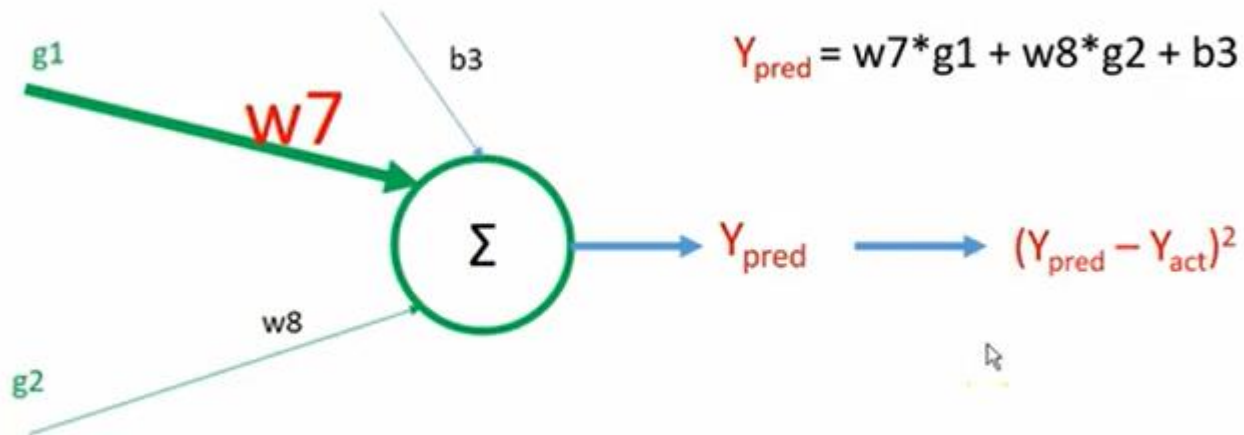
Partial Derivative



$$\frac{\partial Cost}{\partial w7} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial w7}$$

2.5.5 Backpropagation

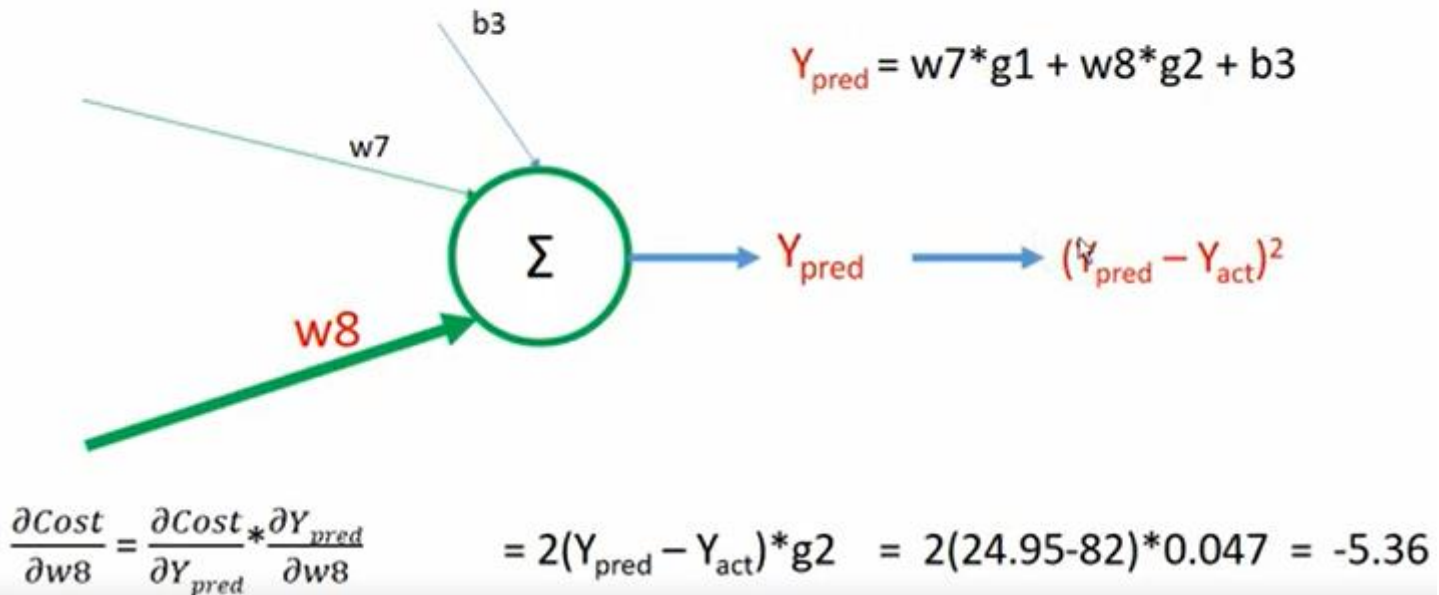
Partial Derivative



$$\frac{\partial Cost}{\partial w7} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial w7} = 2(Y_{pred} - Y_{act}) * g1 = 2(24.95 - 82) * 0.37 = -42.2$$

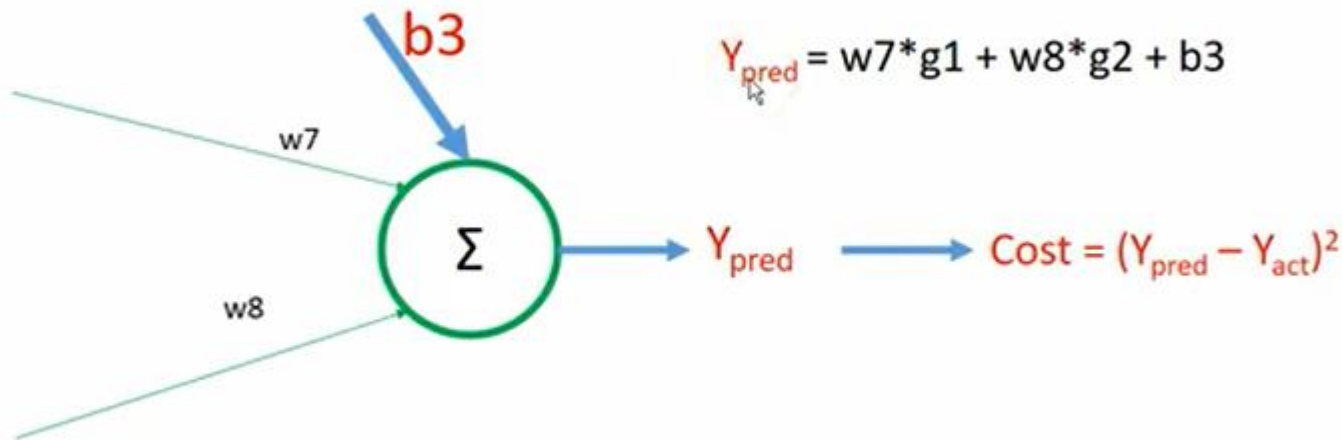
2.5.6 Backpropagation

Partial Derivative



2.5.7 Backpropagation

Partial Derivative



$$\frac{\partial Cost}{\partial b3} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial b3} = 2(Y_{pred} - Y_{act}) * 1 = 2(24.95 - 82) * 1 = -114.1$$

2.5.8 Backpropagation

Update the weights and biases using **gradient descent**.

Walk down the cost function in the direction of the negative partial derivatives, by taking small steps, η , also called the **learning rate**. $\eta = 0.01$

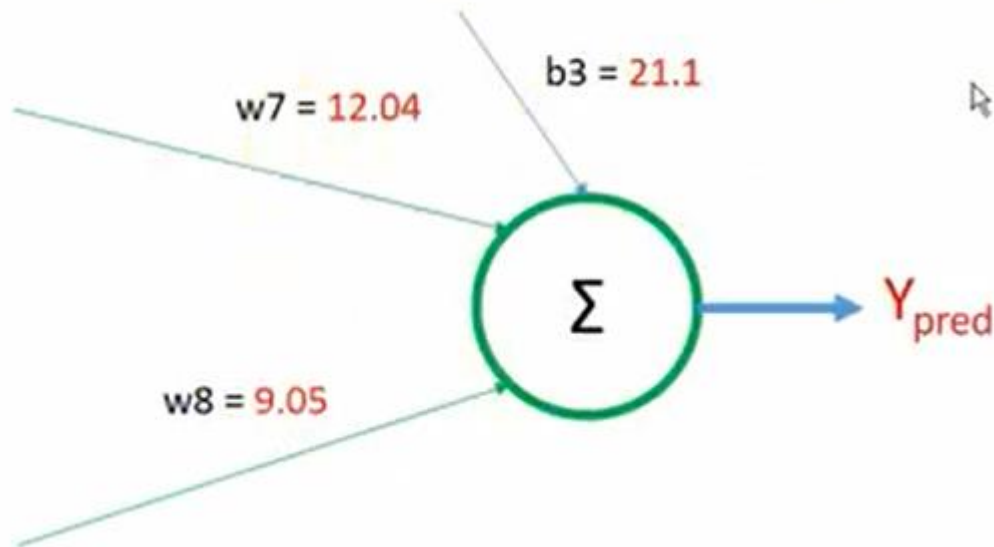
$$w7^+ = w7 - \eta \frac{\partial Cost}{\partial w7} = 12 - 0.01 * (-42.2) = 12.04$$

$$w8^+ = w8 - \eta \frac{\partial Cost}{\partial w8} = 9 - 0.01 * (-5.36) = 9.05$$

$$b3^+ = b3 - \eta \frac{\partial Cost}{\partial b3} = 20 - 0.01 * (-114.1) = 21.1$$

2.5.9 Backpropagation

Updated Weights



2.5.10 Backpropagation

Update Weights and Biases

$$w1^+ = w1 - \eta \frac{\partial Cost}{\partial w1}$$

$$w2^+ = w2 - \eta \frac{\partial Cost}{\partial w2}$$

$$w3^+ = w3 - \eta \frac{\partial Cost}{\partial w3}$$

$$w4^+ = w4 - \eta \frac{\partial Cost}{\partial w4}$$

$$w5^+ = w5 - \eta \frac{\partial Cost}{\partial w5}$$

$$w6^+ = w6 - \eta \frac{\partial Cost}{\partial w6}$$

$$b1^+ = b1 - \eta \frac{\partial Cost}{\partial b1}$$

$$b2^+ = b2 - \eta \frac{\partial Cost}{\partial b2}$$

$$w7^+ = w7 - \eta \frac{\partial Cost}{\partial w7}$$

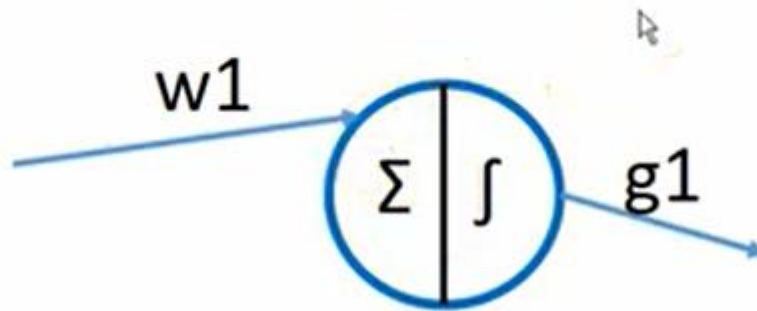
$$w8^+ = w8 - \eta \frac{\partial Cost}{\partial w8}$$

$$b3^+ = b3 - \eta \frac{\partial Cost}{\partial b3}$$

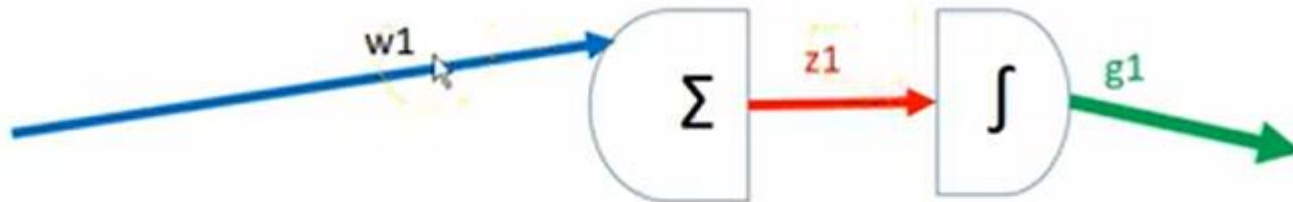
2.5.11 Backpropagation

Let's consider $\frac{\partial g_1}{\partial w_1}$

Remember that there are **two operations** within the neuron...



2.5.12 Backpropagation



$$\frac{\partial g1}{\partial w1} = \frac{\partial g1}{\partial z1} * \frac{\partial z1}{\partial w1}$$

2.5.13 Backpropagation

$$\frac{\partial Cost}{\partial w1} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial g1} * \frac{\partial g1}{\partial z1} * \frac{\partial z1}{\partial w1}$$

$$\frac{\partial Cost}{\partial Y_{pred}} = 2(Y_{pred} - Y_{act})$$

$$(Y_{pred} - Y_{act})^2$$

$$\frac{\partial Y_{pred}}{\partial g1} = w7$$

$$Y_{pred} = w7 * g1 + w8 * g2 + b3$$

$$\frac{\partial g1}{\partial z1} = g1 * (1 - g1) = \left(\frac{1}{1 + e^{-z1}}\right) * \left(1 - \frac{1}{1 + e^{-z1}}\right)$$

$$g1 = \frac{1}{1 + e^{-z1}}$$

$$\frac{\partial z1}{\partial w1} = X1$$

$$z1 = w1 * X1 + w3 * X2 + w5 * X3 + b1$$

2.5.14 Backpropagation

$$\frac{\partial Cost}{\partial w1} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial g1} * \frac{\partial g1}{\partial z1} * \frac{\partial z1}{\partial w1}$$


$$= 2(Y_{pred} - Y_{act}) * w7 * \left[\left(\frac{1}{1 + e^{-z1}} \right) * \left(1 - \frac{1}{1 + e^{-z1}} \right) \right] * X1$$

$$= 2(24.95 - 82) * 12 * \left[\left(\frac{1}{1 + e^{-(-0.5)}} \right) * \left(1 - \frac{1}{1 + e^{-(-0.5)}} \right) \right] * 60$$

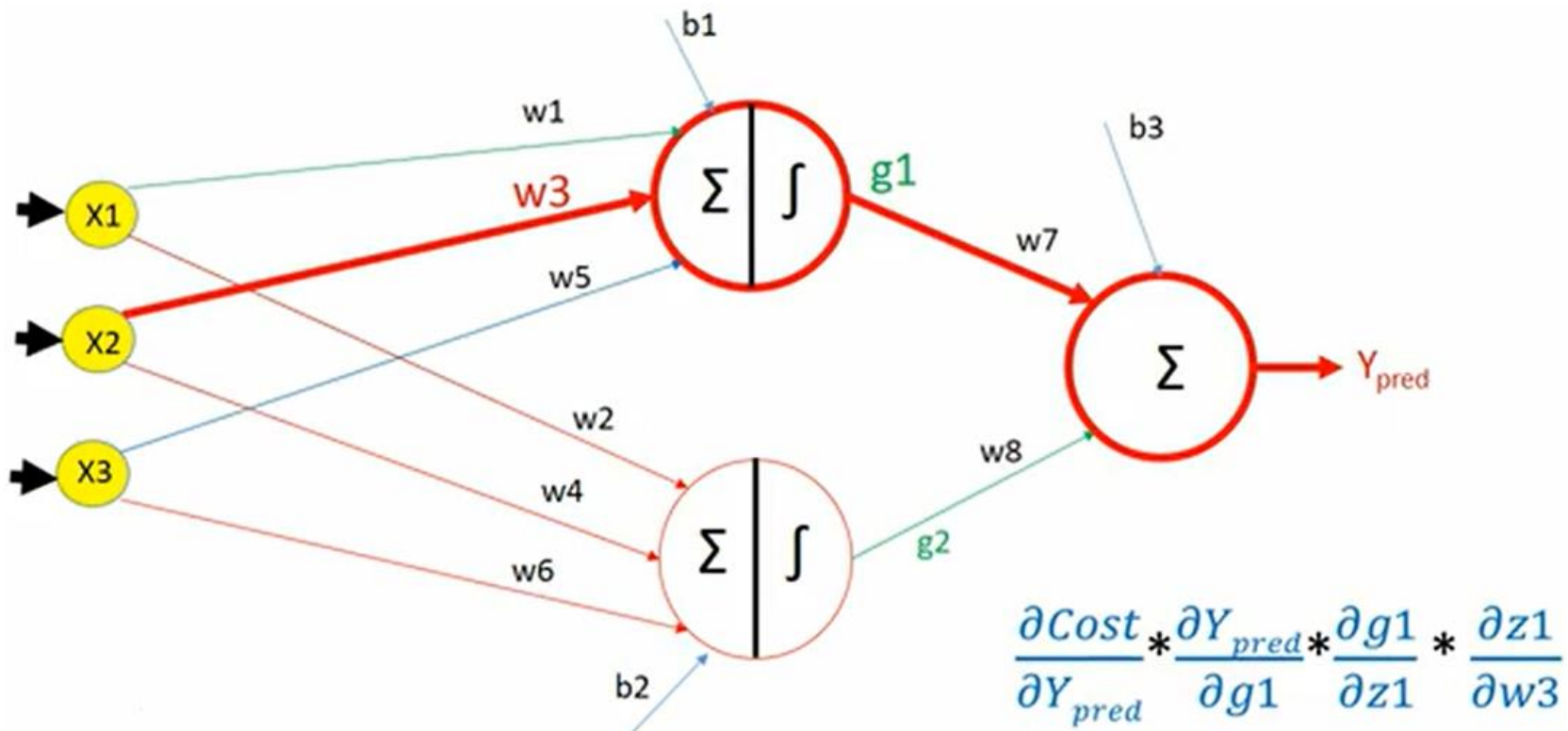
$$= -19,303$$

2.5.15 Backpropagation

Update Weights

$$w1^+ = w1 - \eta \frac{\partial Cost}{\partial w1} = 0.1 - (0.01)*(-19,303) = 193$$

2.5.16 Backpropagation



2.5.17 Backpropagation

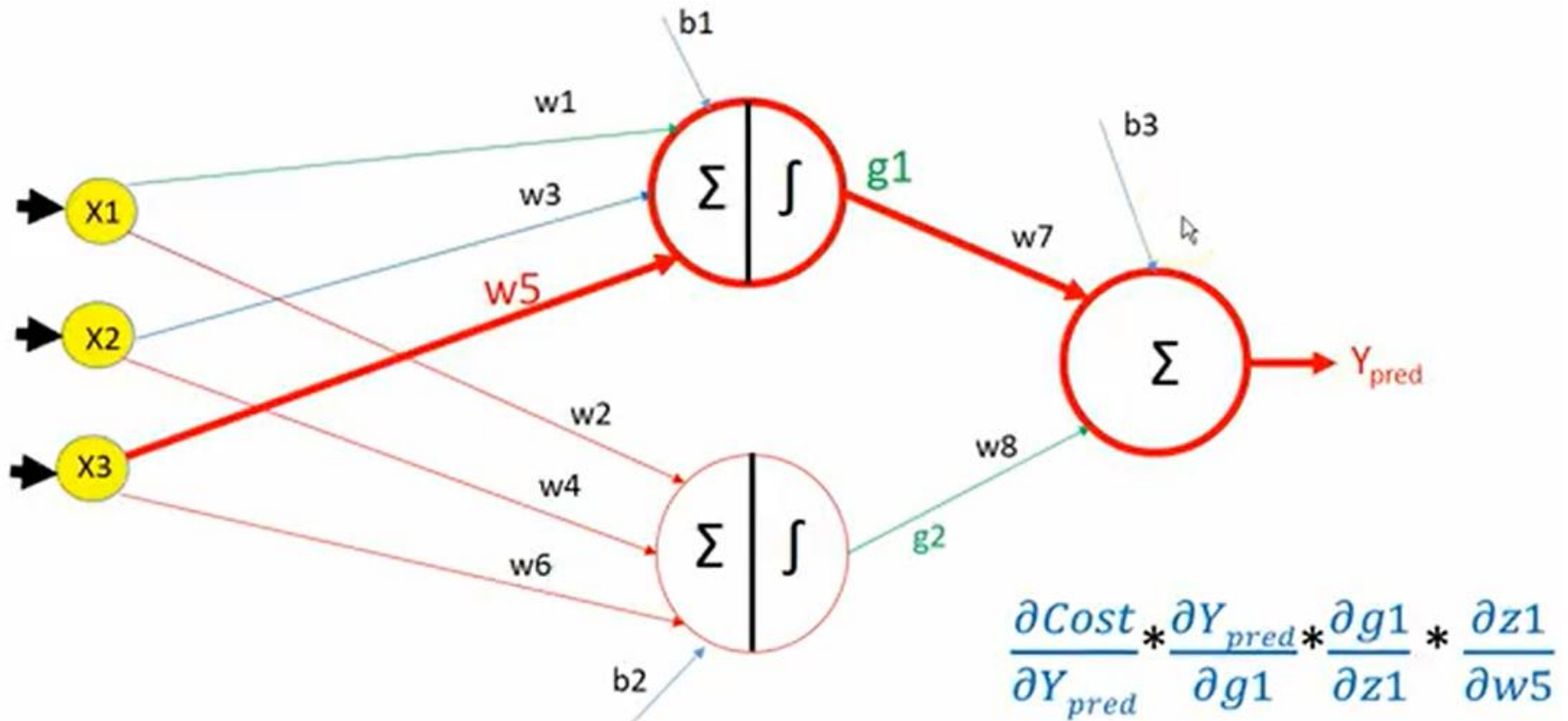
$$\frac{\partial Cost}{\partial w3} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial g1} * \frac{\partial g1}{\partial z1} * \frac{\partial z1}{\partial w3}$$

$$= 2(Y_{pred} - Y_{act}) * w7 * \left[\frac{1}{1 + e^{-z1}} * \left(1 - \frac{1}{1 + e^{-z1}} \right) \right] * X2$$

$$= 2(24.95 - 82) * 12 * \left[\frac{1}{1 + e^{-(-0.5)}} * \left(1 - \frac{1}{1 + e^{-(-0.5)}} \right) \right] * 80$$

$$= -25,738$$

2.5.18 Backpropagation



2.5.19 Backpropagation

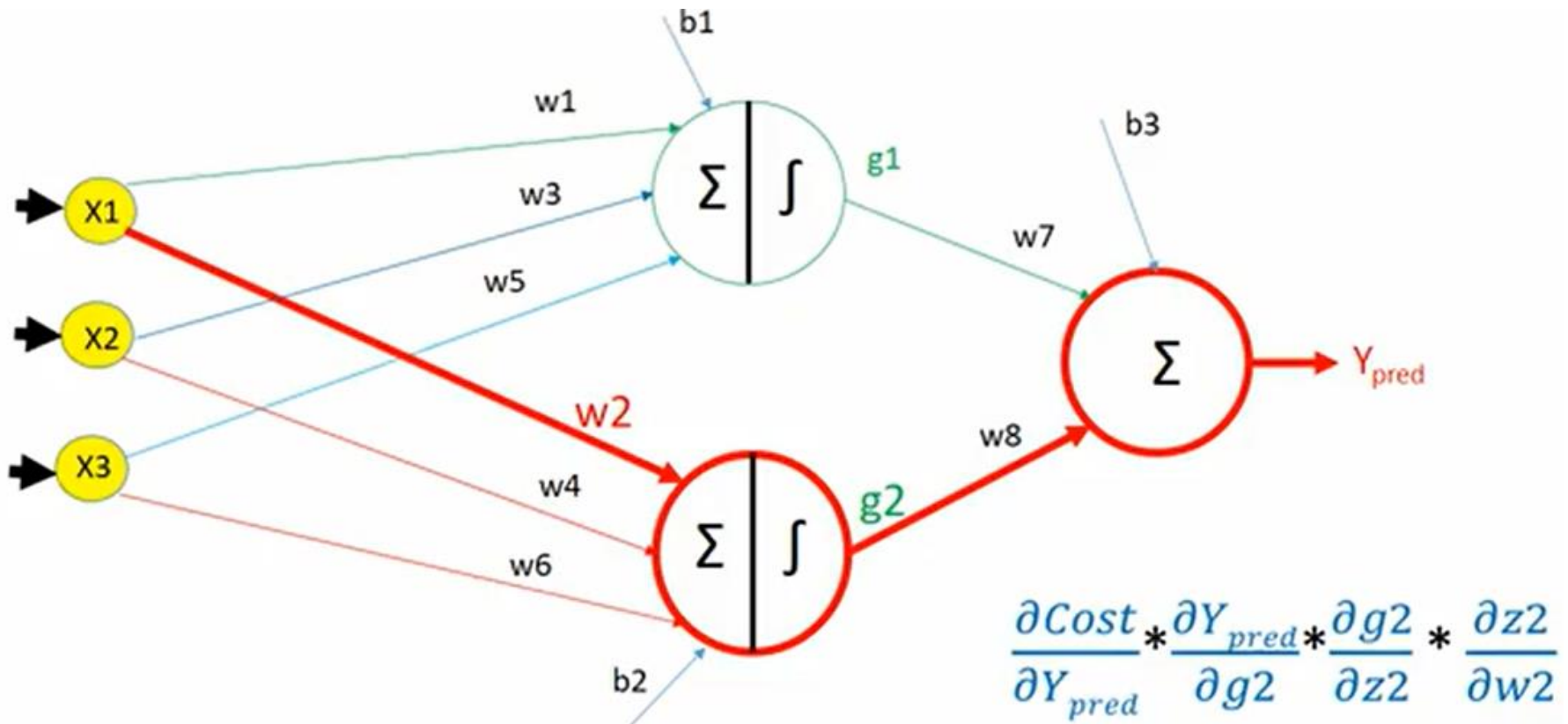
Update Weights

$$w1^+ = w1 - \eta \frac{\partial Cost}{\partial w1} = 0.1 - (0.01)*(-19,303) = 193$$

$$w3^+ = w3 - \eta \frac{\partial Cost}{\partial w3} = 0.1 - (0.01)*(-25,738) = 257.5$$

$$w5^+ = w5 - \eta \frac{\partial Cost}{\partial w5} = 0.1 - (0.01)*(-1608) = 16.2$$

2.5.20 Backpropagation



2.5.21 Backpropagation

$$\frac{\partial Cost}{\partial w2} = \frac{\partial Cost}{\partial Y_{pred}} * \frac{\partial Y_{pred}}{\partial g2} * \frac{\partial g2}{\partial z2} * \frac{\partial z2}{\partial w2}$$


$$= 2(Y_{pred} - Y_{act}) * w8 * \left[\left(\frac{1}{1 + e^{-z2}} \right) * \left(1 - \frac{1}{1 + e^{-z2}} \right) \right] * X1$$

$$= 2(24.95 - 82) * 9 * \left[\frac{1}{1 + e^{-(-3)}} * \left(1 - \frac{1}{1 + e^{-(-3)}} \right) \right] * 60$$

$$= -2,783$$

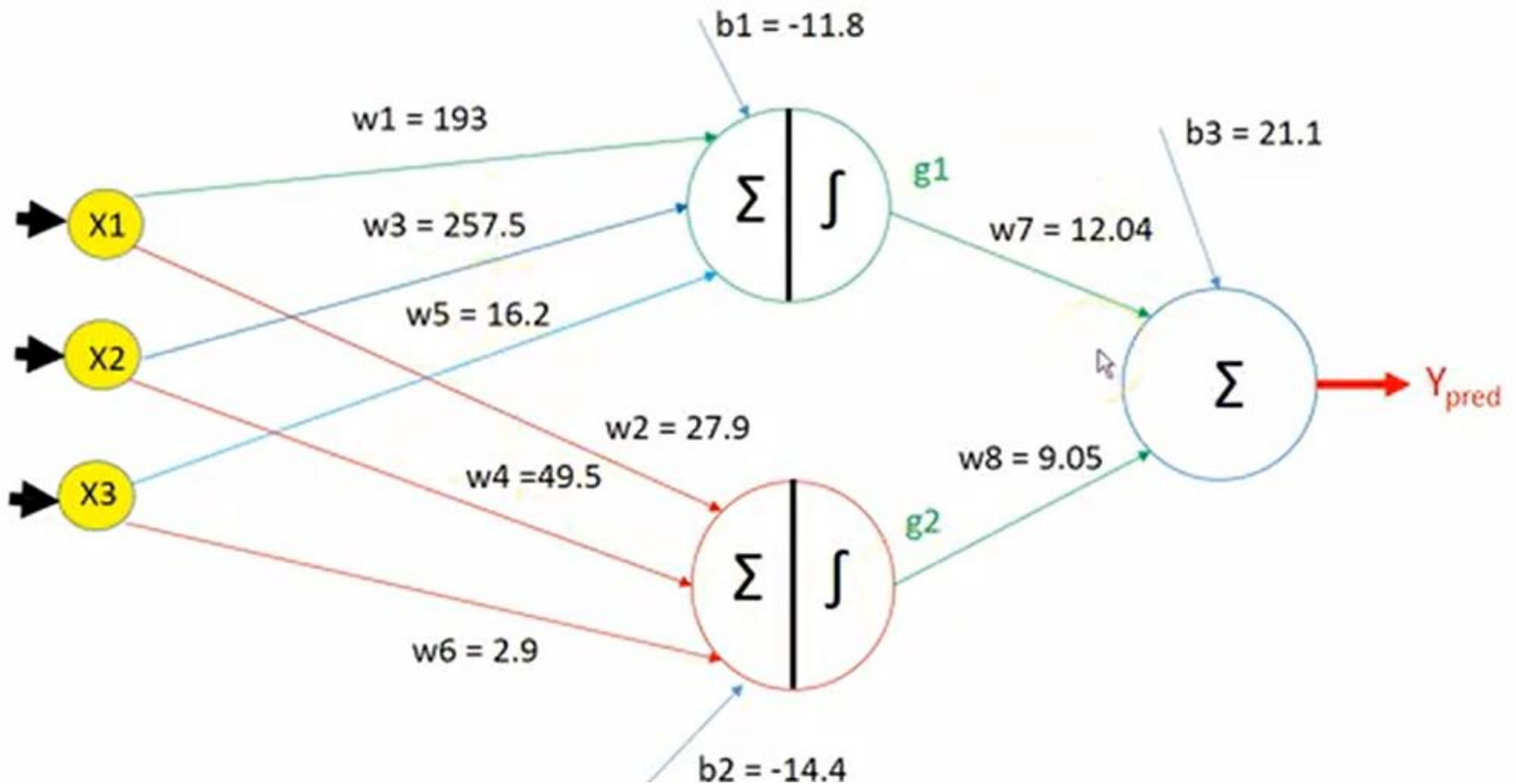
2.5.22 Backpropagation

Update Weights

$$\begin{aligned}w1^+ &= w1 - \eta \frac{\partial Cost}{\partial w1} &= 0.1 - (0.01)*(-19,303) &= 193 \\w2^+ &= w2 - \eta \frac{\partial Cost}{\partial w2} &= 0.15 - (0.01)*(-2783) &= 27.9 \\w3^+ &= w3 - \eta \frac{\partial Cost}{\partial w3} &= 0.1 - (0.01)*(-25,738) &= 257.5 \\b1^+ &= b1 - \eta \frac{\partial Cost}{\partial b1} &= -15 - (0.01)*(-321) &= -11.8 \\w4^+ &= w4 - \eta \frac{\partial Cost}{\partial w4} &= 0.05 - (0.01)*(-3710) &= 37.1 \\w5^+ &= w5 - \eta \frac{\partial Cost}{\partial w5} &= 0.1 - (0.01)*(-1608) &= 16.2 \\w6^+ &= w6 - \eta \frac{\partial Cost}{\partial w6} &= 0.1 - (0.01)*(-309) &= 2.9 \\b2^+ &= b2 - \eta \frac{\partial Cost}{\partial b2} &= -15 - (0.01)*(-61.8) &= -14.4\end{aligned}$$

2.5.23 Backpropagation

Updated Weights & Biases



2.5.24 Backpropagation

- And carry out one more forward pass to calculate the cost function then update the weights through back propagation.
- Then repeat forward propagation using the updated weights and biases from the second propagation pass and the third training example

3.1. Classification of statistical data

- Grouping of data into different categories or classes with similar or homogeneous characteristics
- It includes Geographical, Chronological, Qualitative (Simple and Manifold), and Quantitative or Numerical.
- Geographical Classification
 - The classification of data on the basis of geographical location or region

States	Population (in '000)
Assam	31,205
Bihar	1,04,099
Goa	1,458
Gujarat	60,439
Haryana	25,351

3.1. Classification of statistical data

- Chronological Classification

- The classification of data with respect to different time periods

Year	Number of Students (in '000)
2002	1,349
2007	2,457
2012	2,898
2017	3,145
2022	5,900

3.1. Classification of statistical data

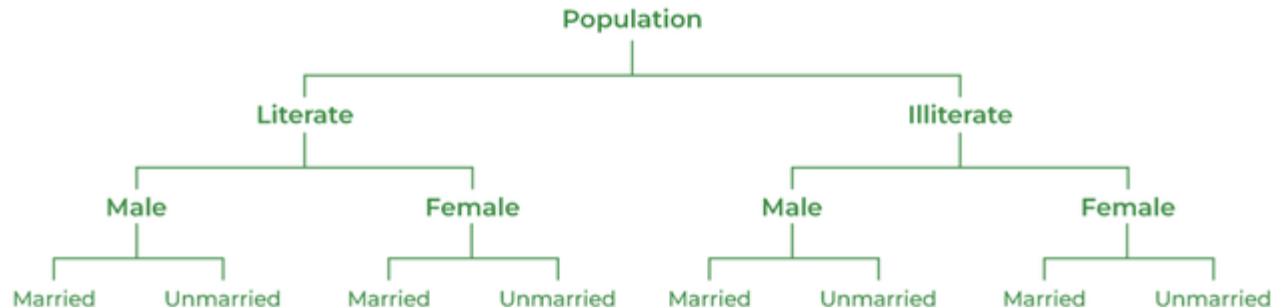
■ Qualitative Classification

- The classification of data on the basis of descriptive or qualitative characteristics like region, caste, sex, gender, education, etc
- Simple Classification
 - When based on only one attribute, the given data is classified into two classes,



■ Manifold Classification

- When based on more than one attribute, the given data is classified into different classes, and then sub-divided into more sub-classes



3.1. Classification of statistical data

■ Quantitative Classification

- The classification of data on the basis of the characteristics, such as age, height, weight, income, etc., that can be measured in quantity

Weight (in kg)	Number of Students
20-25	5
25-30	18
30-35	6
35-40	10
40-45	9

3.2. Measures of Central Tendency & Dispersion

■ Mean

- The sum of all values in a data set divided by the number of values in the set.

Formula:

$$mean = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Consider the data set: 17, 10, 9, 14, 13, 17, 12, 20, 14

$$mean = \frac{\sum x_i}{n} = \frac{17 + 10 + 9 + 14 + 13 + 17 + 12 + 20 + 14}{9} = \frac{126}{9} = 14$$

The mean of this data set is **14**.

3.2. Measures of Central Tendency & Dispersion

■ Median

- The median of a set of data is the “middle element” when the data is arranged in ascending order.
- 1. Put the data in order from smallest to largest.
- 2. Determine the number in the exact center.
 - If there are an odd number of data points, the median will be the number in the absolute middle.
 - If there is an even number of data points, the median is the mean of the two center data points, meaning the two center values should be added together and divided by 2.
- Example:
 - Consider the data set: 17, 10, 9, 14, 13, 17, 12, 20, 14
 - Step 1: Put the data in order from smallest to largest. 9, 10, 12, 13, 14, 14, 17, 17, 20
 - Step 2: Determine the absolute middle of the data. 9, 10, 12, 13, **14**, 14, 17, 17, 20
 - Note: Since the number of data points is odd choose the one in the very middle.
 - The median of this data set is 14.

3.2. Measures of Central Tendency & Dispersion - Mode

- **Mode** is the most frequently occurring measurement in a data set.
 - **One Mode:**
 - If there is only one value that occurs more frequently than any other in the data set, then there is one mode.
 - Example: In the data set {2, 3, 3, 5, 6}, the mode is 3 because it occurs more frequently than any other number.
 - **Multiple Modes:**
 - If more than one value occurs with the same highest frequency in the data set, there are multiple modes.
 - Example: In the data set {2, 3, 3, 5, 5, 6}, both 3 and 5 are modes because they occur with the highest frequency.
 - **No Mode:**
 - If all values in the data set occur with the same frequency (no value occurs more frequently than others), there is no mode.
 - Example: In the data set {2, 3, 5, 6}, there is no mode because each number occurs only once..

3.2. Measures of Central Tendency & Dispersion - Mode

- To determine the mode:
 - 1. Put the data in order from smallest to largest, as you did to find your median.
 - 2. Look for any value that occurs more than once.
 - 3. Determine which of the values from Step 2 occurs most frequently.
- Example:
 - Consider the data set: 17, 10, 9, 14, 13, 17, 12, 20, 14
 - Step 1: Put the data in order from smallest to largest. 9, 10, 12, 13, 14, 14, 17, 17, 20
 - Step 2: Look for any number that occurs more than once. 9, 10, 12, 13, 14, 14, 17, 17, 20
 - Step 3: Determine which of those occur most frequently. 14 and 17 both occur twice.
 - The modes of this data set are 14 and 17.

3.3. Variance and Standard Deviation

- The variance and standard deviation are a measure based on the distance each data value is from the mean.
 - 1. Find the mean of the data.
 - 2. Subtract the mean from each data value (x_i).
 - 3. Square each calculation from Step 2.
 - 4. Add the values of the squares from Step 3.
 - 5. Find the number of data points in your set, called n .
 - 6. Divide the sum from Step 4 by the number n
 - 7. To find the standard deviation, square root this number.

■ Variance = $\frac{\sum(x_i - \bar{x})^2}{n}$

Standard deviation = $\sqrt{\frac{\sum(x_i - \bar{x})^2}{n}}$

3.3. Variance and Standard Deviation

- Example: Calculate the sample variance and sample standard deviation Consider the sample data set: 17, 10, 9, 14, 13, 17, 12, 20, 14.
- Step 1: The mean of the data is 14
- Step 2: Subtract the mean from each data value.
 - $17 - 14 = 3$; $10 - 14 = -4$; $9 - 14 = -5$; $14 - 14 = 0$; $13 - 14 = -1$; $17 - 14 = 3$; $12 - 14 = -2$; $20 - 14 = 6$; $14 - 14 = 0$
- Step 3: Square these values.

$$3^2 = 9; (-4)^2 = 16; (-5)^2 = 25; 0^2 = 0; (-1)^2 = 1; 3^2 = 9; (-2)^2 = 4; 6^2 = 36$$

- Step 4: Add these values together. $9 + 16 + 25 + 0 + 1 + 9 + 4 + 36 = 100$
- Step 5: There are 9 values in our set. Variance = $\frac{100}{9} = 11.11$
- Step 6: Square root this number to find your standard deviation. $= \sqrt{\frac{100}{9}} = 3.33333$

4.1. Correlation

- A statistical concept that describes the relationship between two variables.
- Types of Correlation:
 - **Positive Correlation:**
 - When one variable increases, the other variable tends to increase as well.
 - Example: The more hours you study, the higher your exam scores are likely to be.
 - **Negative Correlation:**
 - When one variable increases, the other variable tends to decrease.
 - Example: The more time you spend working, the lower your free time at home.
 - **No Correlation (Correlation Coefficient Close to 0):**
 - There is no apparent linear relationship between the two variables.
 - Example: The amount of ice cream sold and the number of shark attacks.

4.1. Correlation

■ Correlation Coefficient

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2}}$$

Where:

- X_i and Y_i are individual observations of variables X and Y .
- $\bar{X}$ and $\bar{Y}$ are the means of variables X and Y , respectively.
- The symbol $\sum$ represents the sum across all data points.

4.2. Regression

- A machine learning techniques used to predict a continuous value (dependent variable) based on one or more independent variables (features).
- It tries to build a mathematical model that describes the relationship between the input variables and the output variable.
- Linear Regression: the relationship between the output variable and the input variables is linear.
 - $y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$
 - $x_1, x_2, \dots, x_n$ are the input variables,
 - $b_0, b_1, b_2, \dots, b_n$ are the coefficients to be estimated
- Non-linear Regression: The relationship between the output variable and the input variables is not linear

4.2. Regression - Types of Linear Regression

■ Simple Linear Regression:

- Simple linear regression models the relationship between a single independent variable X and a dependent variable Y using a straight line.
- The equation of a simple linear regression model is typically represented as:
 $Y = \beta_0 + \beta_1 x + \varepsilon$, where β_0 is the intercept, β_1 is the slope, and ε is the error term.

■ Multiple Linear Regression:

- Multiple linear regression models the relationship between two or more independent variables $X_1, X_2, \dots, X_n$ and a dependent variable Y using a linear equation.
- The equation of a multiple linear regression model is typically represented as:
 $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$, where $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ are the coefficients.

4.2. Linear Regression - Formula

$$Y = x\beta + \epsilon$$

We find out the min of $L = (y - x\beta)^2$

$$\Rightarrow L = (y - x\beta)^T (y - x\beta) = (y^T - \beta^T x^T)(y - x\beta)$$

$$\Rightarrow L = y^T y - y^T x\beta - \beta^T x^T y + \beta^T x^T x\beta$$

$$\Rightarrow L = y^T y - (x^T y)^T \beta - \beta^T x^T y + \beta^T x^T x\beta$$

But

$$\partial_{\beta}(a^T \beta) = \partial_{\beta}(\beta^T a) = a$$

$$\frac{\partial L}{\partial \beta} = 0 - x^T y - x^T y + d(\beta^T x^T x) \beta + \beta^T x^T x d(\beta)$$

$$\frac{\partial L}{\partial \beta} = 0 - x^T y - x^T y + x^T x \beta + (\beta^T x^T x)^T$$

$$\frac{\partial L}{\partial \beta} = -x^T y - x^T y + x^T x \beta + x^T x \beta = 0$$

$$\Rightarrow 2x^T x \beta = 2x^T y$$

$$\Rightarrow \beta = (x^T x)^{-1} x^T y$$

- $abs(v) = \sqrt{v v^T}$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$
- $d(uv) = d(u)v + u d(v)$

4.2. Linear Regression - Formula

$$y = X\beta + e$$

We find out the min of $L = (y - X\beta)^2$

$$L = (y - X\beta)^T (y - X\beta)$$

$$L = y^T y - y^T X\beta - (X\beta)^T y + (X\beta)^T (X\beta)$$

$(X\beta)^T y$ is a number so $(X\beta)^T y = y^T X\beta$

$$L = y^T y - (X^T y)^T \beta - \beta^T X^T y + \beta^T X^T X \beta$$

$$\frac{\partial L}{\partial \beta} = 0 - 2X^T y + \frac{\partial(\beta^T X^T X)}{\partial \beta} \beta + \beta^T X^T X \frac{\partial(\beta)}{\partial \beta} = -2y^T X + X^T X \beta + \beta^T X^T X \frac{\partial(\beta)}{\partial \beta}$$

$$\text{But } \frac{\partial(a^T \beta)}{\partial \beta} = \frac{\partial(\beta^T a)}{\partial \beta} = a$$

$$\frac{\partial L}{\partial \beta} = -2X^T y + X^T X \beta + (\beta^T X^T X)^T \frac{\partial(\beta^T)}{\partial \beta} = -2y^T X + X^T X \beta + (\beta^T X^T X)^T$$

$$\frac{\partial L}{\partial \beta} = -2X^T y + 2X^T X \beta = 0, \text{ so } X^T X \beta = X^T y$$

$$\beta = (X^T X)^{-1} X^T y$$

- $\text{abs}(v) = \sqrt{v^T v}$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$
- $d(uv) = d(u)v + u d(v)$

5.1 Probability

- Probability is a measure that quantifies the likelihood or chance of an event occurring
- **Sample Space:** The sample space is the set of all possible outcomes of an experiment. It includes every possible result that could occur.
- **Event:** An event is an outcome or a set of outcomes of an experiment or process.
- **Probability of an Event (P):** The probability of an event is a number between 0 and 1 that represents the likelihood of that event occurring.

$$P(A) = \text{Number of favorable outcomes} / \text{Total number of outcomes} = f/N$$

Where:

$P(A)$ = Probability of an event (event A) occurring

f = Number of ways an event can occur (frequency)

N = Total number of outcomes possible

5.1 Probability – characteristics

- For events A and B such that $A \cap B = \emptyset$,
 - $P(A \cup B) = P(A) + P(B)$
 - $P(A \cap B) = P(A)P(B)$.

5.1 Probability – Conditional probability

- The probability that event A occurs when event B has occurred, denoted $P(A|B)$.

- $P(AB) = \frac{n_{AB}}{n_{\Omega}} \frac{n_A}{n_A} = \frac{n_A}{n_{\Omega}} \frac{n_{AB}}{n_A}$

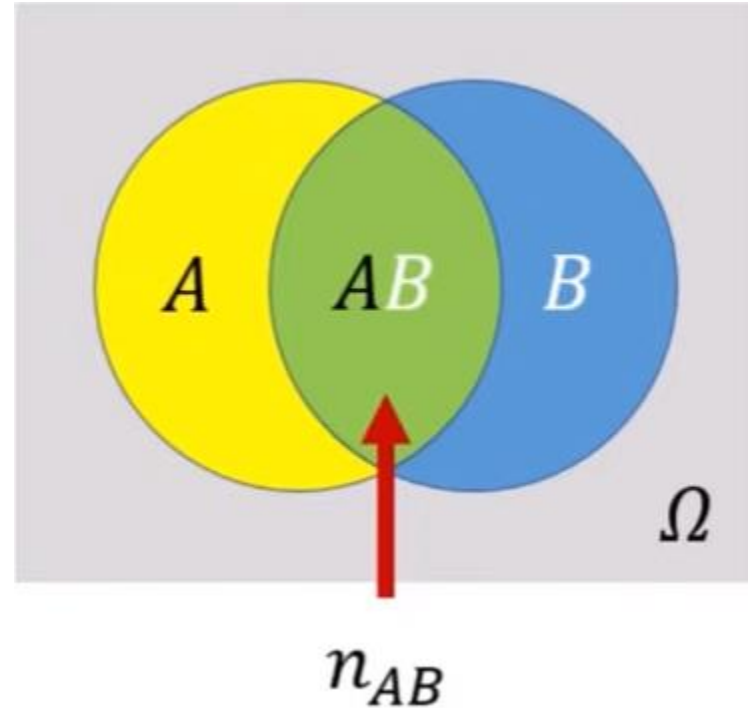
- $P(AB) = P(A) \frac{n_{AB}}{n_A} = P(A)P(B|A)$

- $P(AB) = P(B)P(A|B)$

- $P(B|A) = \frac{P(AB)}{P(A)} = \frac{P(A \cap B)}{P(A)}$

- $P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A \cap B)}{P(B)}$

- $P(AB)$ = Probability of happening of both A and B.

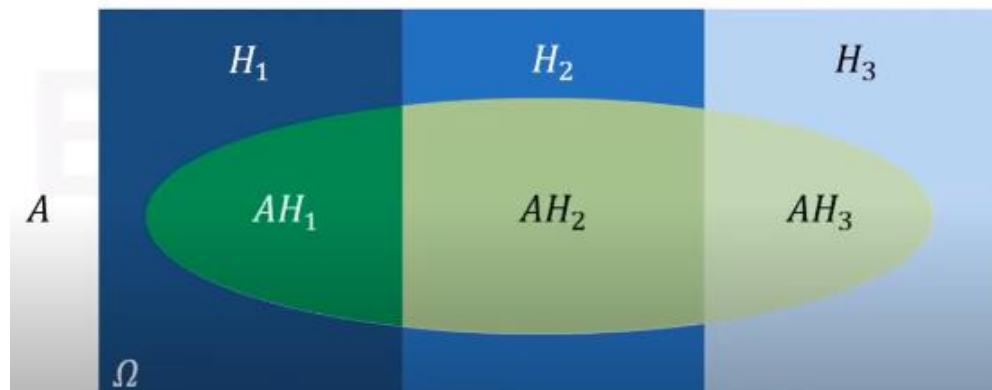


5.1 Probability – Conditional probability

- When a fair die is rolled, what is the probability of A given B where A is the event of getting an odd number and B is the event of getting a number less than or equal to 3?
- **Solution:**
 - To find: $P(A/B)$ using the given information.
 - When a die is rolled, the sample space = $\{1, 2, 3, 4, 5, 6\}$.
 - A is the event of getting an odd number. So $A = \{1, 3, 5\}$.
 - B is the event of getting a number less than or equal to 3. So $B = \{1, 2, 3\}$. $P(B) = 3/6 = 1/2$
 - Then $A \cap B = \{1, 3\}$. So $P(A \cap B) = 2/6 = 1/3$
 - Using the $P(A/B)$ formula:
 - $P(A/B) = P(A \cap B) / P(B)$
 - $P(A/B) = (1/3) / (1/2) = 2/3$
- **Answer:** $P(A/B) = 2 / 3$.

5.1 Probability – Law of Total Probability

- Partition the sample space into mutually exclusive events H_1, H_2, H_3
 - $P(A) = P(AH_1) + P(AH_2) + P(AH_3)$
- We have $P(AB) = P(A|B)P(B)$. So
 - $P(A) = P(A|H_1)P(H_1) + P(A|H_2)P(H_2) + P(A|H_3)P(H_3)$
- Law of Total Probability
 - $P(A) = \sum_{i=1}^n P(A|H_i)P(H_i)$



5.1 Probability – Law of Total Probability

A store sells three types of light bulbs of the same size: 40 white bulbs with a good bulb rate of 99%, 45 yellow bulbs with a good bulb rate of 95%, and 15 green colored bulbs with a good bulb rate of 90%. A customer randomly chooses to buy 1 light bulb from this store. Calculate the probability that this customer will buy a good light bulb?

Solution:

B : customer chooses a good light bulb

A_1, A_2, A_3 : customer chooses white, yellow and blue light bulb

We have $\{A_1, A_2, A_3\}$ as the complete system so:

$$P(B) = P(BA_1) + P(BA_2) + P(BA_3)$$

$$P(B) = P(B|A_1) P(A_1) + P(B|A_2) P(A_2) + P(B|A_3) P(A_3)$$

$$P(B) = 0.99 \cdot 0.4 + 0.95 \cdot 0.45 + 0.9 \cdot 0.15 = 95.85\%$$

5.1 Probability – Bayes' Theorem

- The conditional probability of event A given event B

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$ or $P(A \cap B) = P(A|B)P(B)$

- The conditional probability of event B given event A

- $P(B|A) = \frac{P(B \cap A)}{P(A)}$ or $P(B \cap A) = P(B|A)P(A)$

- We have $P(A \cap B) = P(B \cap A)$.

- $P(A|B)P(B) = P(B|A)P(A)$

- Bayes' theorem

- $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

5.1 Probability – Bayes' Theorem

- 1% of women have cancer.
- If 100 women with cancer undergo mammography, 80 have a +ve result.
- If 100 normal women undergo mammography, 90 have a -ve result.
- If a person has a +ve mammography result, what is the probability that they have cancer, $P(\text{cancer}|\text{+ve}) = ?$

5.1 Probability – Bayes' Theorem

■ Solution:

- We will use the Bayes' Theorem:

$$P(\text{cancer}|\text{+ve}) = \frac{P(\text{+ve}|\text{cancer}) \times P(\text{cancer})}{P(\text{+ve})}$$

- $P(\text{cancer}|\text{+ve})$: probability of a person having cancer given that the test result is positive.
- $P(\text{+ve}|\text{cancer})$: probability of getting a positive test result if the person has cancer = 0.8
- $P(\text{cancer})$: overall probability of a person having cancer = 0.01
- $P(\text{+ve})$: overall probability of getting a positive test result.

- Applying the law of total probability

$$P(\text{+ve}) = P(\text{+ve}|\text{cancer}) \times P(\text{cancer}) + P(\text{+ve}|\text{normal}) \times P(\text{normal})$$

$$P(\text{+ve}) = 0.8 \times 0.01 + (1-0.9) \times (1-0.01) = 0.107$$

- Finally

$$P(\text{cancer}|\text{+ve}) = \frac{0.8 \times 0.01}{0.107} = 0.0747$$

5.2. Probability Distribution

- **Gaussian distribution** (normal distribution or bell curve)
 - is a continuous probability distribution that is characterized by a symmetric bell-shaped curve.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Where:

- $f(x)$ is the probability density function.
- x is the random variable.
- μ is the mean.
- σ is the standard deviation.

5.2. Probability Distribution

■ Uniform Distribution

- The probability density function of a continuous uniform distribution in the interval $[a,b]$ is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b] \\ 0 & \text{if else} \end{cases}$$

- Where:

- $f(x)$ is the probability density function.
- x is a random variable in the interval $[a,b]$.
- a is the lower bound of the interval.
- b is the upper bound of the interval.

- Use Cases of Uniform Distribution:

- Random Number Generation
- Statistical Sampling
- Probability Models

5.2. Probability Distribution

- **Poisson distribution** is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space.
- **Characteristics of the Poisson Distribution:**
 - **Discrete Events:** The Poisson distribution models the number of discrete events that occur within a fixed interval. These events must be independent and occur with a constant mean rate.
 - **Fixed Interval:** The distribution is defined for a fixed interval, and the probability of events occurring is constant across this interval.
 - **Independence:** The occurrence of one event does not affect the occurrence of another. Events are considered independent.
 - **Countable Outcomes:** The random variable in a Poisson distribution represents the count of events, which must be non-negative integers (0, 1, 2, ...).

5.2. Probability Distribution

- The probability mass function of the Poisson distribution is given by: $P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$
- Where:
 - $P(X=k)$ is the probability of observing k events.
 - λ is the average rate of events per interval.
 - k is the number of events (an integer).